Thermal balance and the amplifier entropy photon-number inequality

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Abstract

The entropy photon number is the occupation of a thermal state with a given entropy per mode. We give a variational argument for its sharp lower bound under amplification of two independent finite-energy inputs, allowing correlations within each input group. The proof turns on an agreement of signs: the creation port reverses a logarithmic-mean metric, and thermal relaxation obeys the corresponding signed occupation balance. At a regularized entropy minimum, these two facts cancel the energy corrections and force the thermal-balance defects to vanish. A trace-norm Stein bound, obtained from thermal-attenuator smoothing, then closes the argument. We also give a Gaussian spectral refinement, characterize Gaussian equality by a common symplectic shape, and deduce a sharp correlation bound for the two outputs. The conjecture is due to De Palma, Mari and Giovannetti; the variational framework and four-weight interpolation are adapted from OpenAI's beam-splitter argument.

Acknowledgment and provenance. This manuscript emerged from an iterative, substantially autonomous research process involving several AI systems, with successive rounds of dialogue, mathematical investigation, criticism and rewriting coordinated through human direction. The mathematical development and writing are AI-generated; established contributions are credited in the text.

1 What amplification cannot avoid

Feed vacuum into both ports of a quantum amplifier. The retained output is not vacuum: an amplifier of gain $G > 1$ produces a thermal state with $G - 1$ photons per mode. The pump supplies the energy; the canonical commutation relation fixes this spontaneous contribution. Amplification has an entropy cost even when there is no input uncertainty to amplify.

What happens when the inputs already have entropy? For thermal inputs the answer is immediate: their occupations combine as $Ga + (G-1)(b+1)$. The entropy photon-number inequality asserts that this simple thermal calculation is the sharp lower bound for *every* pair of independent inputs with the same entropies. The inputs may be non-Gaussian, and the modes within either input may be entangled. One scalar per input is nevertheless enough to state the bound.

Let $n \geq 1$ be finite, let $\mathcal{H}_n = \ell^2(\mathbb{N}_0^n)$, and write $N = \sum_{j=1}^n a_j^* a_j$ for total photon number. All logarithms are natural. The entropy of a one-mode thermal state of mean occupation $r \geq 0$ is

$$g(r) = (r+1)\log(r+1) - r\log r, \qquad g(0) = 0.$$

Since g is strictly increasing, an n-mode state has a well-defined *entropy photon number*

$$\mathcal{N}_n(\rho) = g^{-1}\left(\frac{S(\rho)}{n}\right), \qquad S(\rho) = -\operatorname{Tr}\rho\log\rho.$$

This is a thermal entropy coordinate, rather than an energy expectation. Finite energy makes it finite, by $S(\rho) \leq n g(\operatorname{Tr}\rho N/n)$.

Throughout, put $K = G - 1$. The gain-G two-mode squeezing unitary has retained annihilation operators

$$c_j = \sqrt{G}\, a_j + \sqrt{K}\, b_j^*, \qquad 1 \leq j \leq n. \tag{1.1}$$

For the product input $\rho_A \otimes \rho_B$, denote the retained state by $\Gamma_G(\rho_A, \rho_B)$. The product is between the two groups; there is no product assumption within a group.

Theorem 1.1 (Amplifier entropy photon-number inequality)**.** *For every finite $n \geq 1$, every $G > 1$, and independent finite-energy n-mode inputs,*

$$\mathcal{N}_n\big(\Gamma_G(\rho_A, \rho_B)\big) \geq G\mathcal{N}_n(\rho_A) + K\big(\mathcal{N}_n(\rho_B) + 1\big). \tag{1.2}$$

Products of thermal states with a common mean within each group, and their coherent displacements, attain equality.

The extra unit in (1.2) is the identity in $bb^* = b^*b + I$. In particular, the thermal reference output has mean

$$R = Ga + K(b + 1). \tag{1.3}$$

De Palma, Mari and Giovannetti formulated this amplifier conjecture [1]. The proof here adapts the regularized variational method and four-weight interpolation of OpenAI's passive, beam-splitter argument [2]. Its active part is the agreement between a reversed thermal metric and a signed relaxation balance.

The argument before the machinery

Suppose the bound fails. Match thermal references to the input entropies and subtract their entropy tangent. The resulting functional has a negative value. We modify it just enough to attain a minimum at states for which differentiation is legitimate: a downstream thermal port supplies strict comparison slack, thermal replacement controls the output logarithm, and a cubic moment penalty controls the input domains. Only the last penalty will tend to zero.

At the minimum, entropy curvature gives a logarithmic-mean comparison. The two thermal occupation balances turn it into two weighted comparisons; independence supplies two arithmetic variance comparisons. These are the four endpoints of an interpolation argument. The interpolated metric reverses orientation at the creation port B.

Now relax each input toward its own thermal reference. The amplifier carries this evolution to the corresponding output relaxation precisely because the reference occupations obey the signed balance in (1.3). Entropy production in this relaxation is a squared thermal-balance defect minus a centered-energy correction. Pairing the stationary equations with the relaxation cancels all those corrections.

Equation (6.5) is where the two parts of the proof meet. The metric estimate bounds the output defect by the input defects; stationarity gives the input defects a strictly larger coefficient. They must vanish as the cubic penalty is removed. A thermal Stein estimate then forces the minimizing inputs back to their thermal references, where the entropy tangent is zero. The negative minimum has nowhere to go.

Sections 3–6 develop this argument, and Section 7 completes it. The Stein estimate is also an independent statement about approximate thermal balance. Section 8 gives a separate

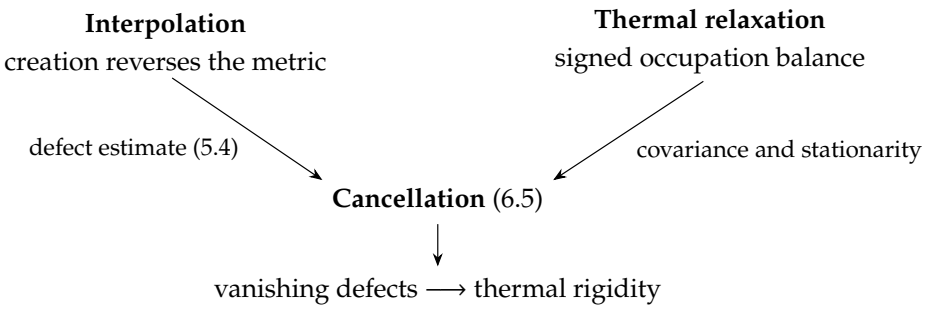


Figure 1: The two signs meet at the variational minimum. Covariant relaxation gives the cancellation identity; interpolation controls the remaining defects. Strict comparison slack forces those defects to vanish as the cubic penalty is removed.

Gaussian proof and identifies its equality geometry. Section 9 explains what the two marginal bounds say about correlations created by amplification.

The surrounding results

The passive conjecture originated with Guha, Erkmen and Shapiro [3]. The quantum entropy-power inequalities of König–Smith and De Palma–Mari–Giovannetti [1, 4] control an exponential of entropy; in general they do not recover the sharp value for unequal thermal inputs. The one-mode constrained Gaussian optimizer [5] and its product-basis diagonal multimode extension [6] settle substantial restricted cases. They do not allow two freely varying arbitrary multimode inputs.

Several established tools clarify different parts of the present proof. The shifted logarithmic means are the detailed-balance metrics of Carlen–Maas [7]; Gao–Rouzé's beam-splitter smoothing [8] supplies the analytic mechanism behind the Stein estimate; and Bhatia–Jain's symplectic variational principle [9] makes the Gaussian equality argument transparent. We indicate where each enters. The bosonic domains and the active-port identities still require their own justifications.

2 Entropy curvature in thermal coordinates

Fix positive reference means and write

$$h(r) = g'(r) = \log(1 + r^{-1}), \qquad \tau_r = (r+1)^{-n} e^{-h(r)N}.$$

The factor $1/h(r)$ converts an entropy-per-mode variation into an occupation variation. For the thermal entropy function

$$F_G(s,t) = g\big(Gg^{-1}(s) + K\big(g^{-1}(t) + 1\big)\big),$$

put $r_0 = Gr_A + K(r_B + 1)$ and $h_s = h(r_s)$, for $s = 0, A, B$. At $(s,t) = (g(r_A), g(r_B))$,

$$\partial_s F_G = \frac{Gh_0}{h_A}, \qquad \partial_t F_G = \frac{Kh_0}{h_B}. \tag{2.1}$$

Thus the natural entropy tangent is

$$\frac{S(\rho_C) - ng(r_0)}{h_0} - \frac{G}{h_A}\big(S(\rho_A) - ng(r_A)\big) - \frac{K}{h_B}\big(S(\rho_B) - ng(r_B)\big).$$

When the references match the input entropies, a strict violation of (1.2) makes this expression negative. We use this local tangent property; no global supporting-plane claim is needed.

Entropy curvature introduces the logarithmic mean

$$\ell(p,q) = \frac{p-q}{\log p - \log q}, \qquad \ell(p,p) = p.$$

For a faithful state $\rho = \mathrm{diag}(p_l)$, the relative-entropy Hessian has coordinate weights $1/\ell(p_l, p_k)$. Its dual therefore has weights $\ell(p_l, p_k)$. The thermal rates enter through the balance identity

$$(r+1)\tau_r a_j^* = r a_j^* \tau_r,$$

which follows from the Gibbs form and $Na_j^* = a_j^*(N+1)$. We shall need the curvature form with its two spectral arguments shifted by these rates. On centered observables, $\operatorname{Tr}\rho Z = 0$, use the normalization

$$B_{r,\rho}(Z) = h(r)\sum_{l,k}\ell(p_l,p_k)|Z_{lk}|^2, \tag{2.2}$$

$$M^{\downarrow}_{r,\rho}(Z) = h(r)\sum_{l,k}\ell(rp_l,(r+1)p_k)|Z_{lk}|^2, \tag{2.3}$$

$$M^{\uparrow}_{r,\rho}(Z) = h(r)\sum_{l,k}\ell((r+1)p_l,rp_k)|Z_{lk}|^2. \tag{2.4}$$

The first form is the normalized Bogoliubov–Kubo–Mori form. The other two insert the thermal downward and upward rates into its two spectral arguments.

This small change of weights is the geometric part of the proof. The ordered pair $(r, r+1)$ has an orientation: adjunction changes it to $(r+1, r)$. More precisely,

$$M^{\uparrow}_{r,\rho}(Z) = M^{\downarrow}_{r,\rho}(Z^*).$$

Monotonicity and homogeneity of the logarithmic mean also give

$$rB_{r,\rho}(Z) \le M^{\diamond}_{r,\rho}(Z) \le (r+1)B_{r,\rho}(Z), \qquad \diamond \in \{\downarrow,\uparrow\}. \tag{2.5}$$

These are the frequency-shifted logarithmic means in the detailed-balance geometry of Carlen–Maas [7, Definition 5.7 and Lemma 5.8]; adjunction reverses the modular frequency. That finite-dimensional framework explains which metric to expect. Appendices A and B supply the infinite-dimensional estimates used here. The hyperbolic coordinates needed to prove interpolation appear only in Appendix C.

3 Giving a negative minimum enough room

The variational argument must do two jobs at once: retain a strict violation and make its minimum regular enough to test against an unbounded relaxation generator. Three modifications serve distinct purposes. The auxiliary attenuator creates slack in both occupation balances. Thermal replacement bounds the output logarithm. The cubic penalty supplies the moments needed for stationarity. We choose them in that order.

Assume (1.2) has a strict counterexample. We may first give it finite number support. Indeed, finite-mode Gaussian unitaries have a linear energy bound, while energy balls are trace-norm compact and entropy is continuous on them; normalized number cutoffs therefore preserve a strict violation. Choose positive r_A, r_B close to the entropy photon numbers of this witness. A zero input entropy causes no difficulty: $g(r)/h(r) \to 0$ as $r \downarrow 0$, while the output reference remains at least $K > 0$. Appendix A gives the cutoff and continuity details.

Slack from a thermal port

Mix the amplifier output with an independent thermal state of mean $r_D = r_0 = Gr_A + K(r_B+1)$. Let $u > 0$ be the small transmission from this added port, and put

$$\lambda_A = (1-u)G, \qquad \lambda_B = (1-u)K, \qquad \lambda_D = u.$$

Call the retained state γ. Its annihilation operator has one creation port, so $\lambda_A - \lambda_B + \lambda_D = 1$. The two corresponding occupation identities are

$$r_0 = \lambda_A r_A + \lambda_B(r_B+1) + \lambda_D r_D, \tag{3.1}$$

$$r_0 + 1 = \lambda_A(r_A+1) + \lambda_B r_B + \lambda_D(r_D+1). \tag{3.2}$$

The added port contributes positively to both lines. Omitting its contribution leaves room for a common factor $1+\varepsilon$: any $0 < \varepsilon \le u/(1-u)$ satisfies

$$(1+\varepsilon)\big[\lambda_A r_A + \lambda_B(r_B+1)\big] \le r_0, \tag{3.3}$$

$$(1+\varepsilon)\big[\lambda_A(r_A+1) + \lambda_B r_B\big] \le r_0 + 1. \tag{3.4}$$

This is why the auxiliary state is thermal. It supplies comparison slack while contributing no thermal-balance defect of its own.

For small u, strong continuity of the downstream beam splitter, its uniform energy bound, and entropy continuity preserve the negative entropy tangent. Choose $\varepsilon > 0$ also small enough that adding the witness's finite relative-entropy penalty preserves negativity. Both occupation slacks are strict at $\varepsilon = 0$.

Control of logarithms and moments

Replace a small fraction of the output by its thermal reference:

$$\rho_0 = (1-\theta)\gamma + \theta\tau_{r_0}, \qquad w_i = (1-\theta)\lambda_i \quad (i = A, B). \tag{3.5}$$

Choose $\theta > 0$ small enough to retain the penalized negative witness and to satisfy

$$\theta \le \frac{\varepsilon^2 h_0 r_0}{4(\varepsilon+2)}. \tag{3.6}$$

This replacement gives a faithful output with a logarithm bounded by a linear number operator. It also introduces a small centering error; (3.6) is chosen to absorb that error later.

All these parameters are now fixed. With $W_s = 1 + N_s$, minimize, for $\zeta > 0$,

$$\begin{aligned} J_\zeta(\rho_A, \rho_B) = {} & \frac{S(\rho_0) - n g(r_0)}{h_0} - \sum_{i=A,B} \frac{w_i}{h_i}\big[S(\rho_i) - n g(r_i)\big] \\ & + \varepsilon \sum_{i=A,B} \frac{w_i}{h_i} D(\rho_i \| \tau_{r_i}) + \zeta \sum_{i=A,B} \operatorname{Tr} \rho_i W_i^3. \end{aligned} \tag{3.7}$$

Here $D(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$. At the fixed finite-support witness, the cubic term tends to zero. Hence there is a fixed $c > 0$ such that, for every sufficiently small ζ,

$$\inf J_\zeta \le -c. \tag{3.8}$$

Expanding the thermal relative entropy writes the objective, up to fixed constants, as

$$\frac{S(\rho_0)}{h_0} - \sum_i \frac{(1+\varepsilon)w_i}{h_i} S(\rho_i) + \varepsilon \sum_i w_i \operatorname{Tr}\rho_i N_i + \zeta \sum_i \operatorname{Tr}\rho_i W_i^3.$$

The linear energy term dominates the logarithmic entropy bound. Upper sublevels therefore have uniformly bounded energy, independently of ζ; at each fixed $\zeta > 0$ they also have bounded third moment. Compactness, entropy continuity and lower semicontinuity give an attained minimum.

At a minimum put $H_s = -\log\rho_s$. Thermal replacement gives $0 \le H_0 \le CW_0$ as quadratic forms. Let T_i be the positive unital slice of an output observable against the fixed other ports, so that $\operatorname{Tr}\rho_i T_i Z = \operatorname{Tr}\gamma Z$. The Gaussian energy bound then gives $0 \le T_i H_0 \le C_i W_i$.

Replace output entropy by its supporting cross entropy against H_0. This majorizes J_ζ and agrees with it at the minimum. The Gibbs variational principle identifies each minimizing input as the Gibbs state of the form Hamiltonian

$$\begin{aligned} K_i &= \frac{h_i}{(1+\varepsilon)w_i}\left[\frac{1-\theta}{h_0} T_i H_0 + \varepsilon w_i N_i + \zeta W_i^3\right] \\ &= k_i W_i^3 + W_i^{1/2} B_i W_i^{1/2}, \qquad k_i > 0, \quad B_i \ge 0 \text{ bounded}. \end{aligned}$$

Its form domain is $D(W_i^{3/2})$ and its resolvent is compact. The useful gain in regularity is seen directly from an eigenvector $K_i\psi = v\psi$:

$$k_i W_i^{5/2}\psi = vW_i^{-1/2}\psi - B_i W_i^{1/2}\psi.$$

Together with polynomial eigenvalue counting, this proves faithfulness and a finite fifth number moment for the Gibbs state. Thus $0 \le H_i \le C_i W_i^3$, and the exact Euler–Lagrange identity is

$$\frac{(1+\varepsilon)w_i}{h_i}H_i = \frac{1-\theta}{h_0}T_iH_0 + \varepsilon w_i N_i + \zeta W_i^3 + c_i I. \tag{3.9}$$

The cubic term is there for the domains. A fifth moment propagates through the fixed Gaussian unitary and permits the generator–logarithm pairing used below. Appendix A proves these assertions as form and weighted trace-class statements; no identification with a merely formal operator sum is required.

4 Four comparisons and one reversed metric

At the regularized minimum, two kinds of comparison are available. Entropy curvature controls the logarithmic mean; independence controls the two arithmetic variances. The occupation slacks turn the first comparison into two endpoint inequalities. Interpolation assembles these four endpoints into the thermal metric.

Use subscripts $s = 0, A, B$ for the forms (2.2)–(2.4) at the minimizing states. For output tests centered in ρ_0, set

$$L_i Z = \frac{T_i Z - \mathrm{Tr}(\gamma Z)I}{\sqrt{\lambda_i}}.$$

The centering is with respect to γ inside the slice, and with respect to ρ_0 at the source.

The endpoints

With the other input fixed, (3.9) cancels all linear terms in a variation:

$$J_\zeta(\widetilde{\rho}_i) - J_\zeta(\rho_i) = \frac{(1+\varepsilon)w_i}{h_i}D(\widetilde{\rho}_i\|\rho_i) - \frac{1}{h_0}D(\widetilde{\rho}_0\|\rho_0) \ge 0. \tag{4.1}$$

For a traceless Hermitian finite-eigenblock perturbation A, the input Hessian has weights $1/\ell(p_{i,l}, p_{i,k})$. Test the output relative entropy by its Gibbs variational formula against a centered finite-eigenblock observable Z. Optimizing the resulting quadratic inequality over A yields

$$B_i(L_i Z) \le \frac{1+\varepsilon}{1-\theta}B_0(Z). \tag{4.2}$$

The trace-zero approximation corrects the trace in one fixed diagonal entry; Hermitian and anti-Hermitian parts give the complex version. Lemma A.4 supplies the full argument without differentiating output entropy in an unbounded direction.

The arithmetic comparisons use only independence. Write, on centered tests,

$$A_s^+(Z) = \mathrm{Tr}\,\rho_s Z Z^*, \qquad A_s^-(Z) = \mathrm{Tr}\,\rho_s Z^* Z.$$

Centered observables on distinct input ports are orthogonal in both inner products. Projection onto the one-port subspaces, followed by the replacement mixture, gives

$$\sum_i w_i A_i^\pm(L_i Z) \le A_0^\pm(Z). \tag{4.3}$$

Finally, (4.2) and the two occupation slacks give

$$w_A r_A B_A(L_A Z) + w_B(r_B + 1)B_B(L_B Z) \le r_0 B_0(Z), \tag{4.4}$$
$$w_A(r_A + 1)B_A(L_A Z) + w_B r_B B_B(L_B Z) \le (r_0 + 1)B_0(Z). \tag{4.5}$$

The B-port rates occur in the opposite order. This is the same extra unit that appeared in the thermal output occupation.

What interpolation transfers

For coordinate weights $X_s > 0$, let

$$Q_{X,s}(Z) = \sum_\alpha X_s(\alpha)|Z_\alpha|^2.$$

Say that a family $X = (X_s)$ satisfies *comparison* when $Q_{X,0}(Z) \ge \sum_i w_i Q_{X,i}(L_i Z)$ on the source test space. The needed interpolation statement has a simple logarithmic-mean form.

Lemma 4.1 (Twisted logarithmic-mean transfer)**.** *At every coordinate of every space let $p, q, a, b > 0$. Assume every source test has finitely many values of $(\log(p/q), \log(a/b))$, the source space is invariant under bounded scalar functions of this pair, and its four endpoint forms are finite. If comparison holds for*

$$p, \qquad q, \qquad \frac{a\ell(p,q)}{\ell(a,b)}, \qquad \frac{b\ell(p,q)}{\ell(a,b)},$$

then it holds for $\ell(ap, bq)/\ell(a, b)$.

Proof. Absorb $\ell(p, q)$ into the fixed coordinate measure and put $x = \frac{1}{2}\log(p/q)$, $y = \frac{1}{2}\log(a/b)$. For $f(t) = t/\sinh t$, with $f(0) = 1$, the normalized endpoints are $f(x)e^{\pm x}$ and $f(y)e^{\pm y}$; the target is $f(x)f(y)/f(x + y)$. Apply Theorem C.1 with $m = 1$. On the finitely many parameter values of a source test, any scalar multiplier agrees with a bounded one, so the stated invariance suffices. □

The lemma is an algebraic rewriting of OpenAI's four-weight theorem [2]. Its final step is variational: an optimal splitting of the source test remains an admissible splitting after a linear slice, and each image can only improve its quadratic minimum by choosing its own splitting. The analytic work is to construct the positive auxiliary weight that makes this minimum the desired logarithmic mean. Appendix C does so with parallel sums and a Stieltjes representation, and gives the full proof.

In our application $p = p_l$ and $q = p_k$. Choose

$$(a_0, b_0) = (r_0, r_0 + 1), \qquad (a_A, b_A) = (r_A, r_A + 1), \qquad (a_B, b_B) = (r_B + 1, r_B).$$

Since $\ell(r, r + 1) = 1/h(r)$, the arithmetic endpoints are (4.3), and the other two are (4.4)–(4.5). The result is

$$w_A M_A^{\downarrow}(L_A Z) + w_B M_B^{\uparrow}(L_B Z) \le M_0^{\downarrow}(Z). \tag{4.6}$$

The creation port has reversed the metric.

One test-space detail matters. A centered finite-eigenblock observable is a finite block minus its state mean times I, so it may have an infinite identity tail. It still has only finitely many values of $\log(p_l/p_k)$. Every diagonal entry, including the tail, has parameter zero; a parameter multiplier rescales all these entries by the same constant and preserves centering. This verifies the source invariance required by the lemma. It imposes no commutation between a conditional slice and the frequency multipliers. The bounds (2.5) and (4.2) extend (4.6) to the completed centered spaces and to bounded observables.

5 The same balance in the dynamics

The metric comparison becomes useful when applied to a functional that detects thermal balance. We track coherent displacement separately from the centered defect. For $s = 0, A, B$, put

$$\mu_{s,j} = \operatorname{Tr} \rho_s a_{s,j}, \qquad d_{s,j} = a_{s,j} - \mu_{s,j} I, \qquad e'_s = \operatorname{Tr} \rho_s N_s - |\mu_s|^2, \qquad |\mu_s|^2 = \sum_j |\mu_{s,j}|^2.$$

The two centered defects are

$$q^{\downarrow}_{s,j}(Z) = (r_s + 1) \operatorname{Tr} \rho_s d^*_{s,j} Z - r_s \operatorname{Tr} \rho_s Z d^*_{s,j}, \tag{5.1}$$

$$q^{\uparrow}_{s,j}(Z) = r_s \operatorname{Tr} \rho_s d_{s,j} Z - (r_s + 1) \operatorname{Tr} \rho_s Z d_{s,j}. \tag{5.2}$$

Both defects vanish at the reference τ_{r_s}. Finite energy defines these pairings for bounded Z by Hilbert–Schmidt factorization. They annihilate I, and adjunction gives

$$q^{\uparrow}(Z) = -\overline{q^{\downarrow}(Z^*)}, \qquad \|q^{\uparrow}\|_{M^{\uparrow},*} = \|q^{\downarrow}\|_{M^{\downarrow},*}.$$

The conjugation keeps both functionals complex-linear. Here $\|q\|_{M,*} = \sup_{M(Z)\le 1} |q(Z)|$ on the centered test space. For a vector of defects we sum the squared dual norms over modes.

How the defects pass through the amplifier

The Heisenberg transformation and the occupation balances give

$$q^{\downarrow}_0 = w_A q^{\downarrow}_A \circ L_A - w_B q^{\uparrow}_B \circ L_B + \theta(1-\theta)\overline{\mu_\gamma} \operatorname{Tr}[(\gamma - \tau_{r_0}) \cdot], \tag{5.3}$$

where $\mu_{\gamma,j} = \operatorname{Tr} \gamma c_j$. This is a mode-indexed identity. It contains the reversal from (4.6), together with a minus sign at the creation port.

To see why no additional commutator survives, pull a retained test through the unitary. The coefficient left after subtracting the input defects is

$$r_0(\lambda_A - \lambda_B + \lambda_D) - \lambda_A r_A - \lambda_B(r_B + 1) - \lambda_D r_D = 0.$$

The auxiliary thermal port has zero defect. The last term of (5.3) is the centering error of the replacement mixture. For unbounded ladders the commutation with discarded modes is justified by cutting off a discarded operator in *its own* number basis, where it still commutes with every retained bounded test. Finite-energy factorization then passes to the limit. A cutoff in the joint number would destroy that commutation. Appendix B gives the precise weak identity.

Introduce the two total input defects

$$A_*^2 = \sum_i w_i \sum_j \|q^{\downarrow}_{i,j}\|^2_{M_i^{\downarrow},*}, \qquad M_*^2 = \sum_i w_i |\mu_i|^2.$$

The first measures centered thermal imbalance; the second records displacement. By (4.6), the first two terms in (5.3) have dual norm at most A_*. For the remaining term, the scalar mixture map

$$J(Z) = \sqrt{\theta(1-\theta)}\, \operatorname{Tr}[(\gamma - \tau_{r_0}) Z]$$

is bounded by either arithmetic variance. Interpolating their logarithmic mean gives $|J(Z)|^2 \le B_0(Z)/h_0$, hence $|J(Z)|^2 \le M_0^{\downarrow}(Z)/(h_0 r_0)$. Also $(1-\theta)|\mu_\gamma|^2 \le 2M_*^2$. Thus

$$\|q^{\downarrow}_0\|_{M^{\downarrow}_0,*} \le A_* + \sqrt{\frac{2\theta}{h_0 r_0}}\, M_*.$$

Young's inequality and (3.6) yield

$$\|q^{\downarrow}_0\|^2_{M^{\downarrow}_0,*} \le A_*^2 + \frac{\varepsilon}{2}(A_*^2 + M_*^2). \tag{5.4}$$

Relaxation with the right reference

Consider the uncentered thermal relaxation generator

$$\mathcal{L}_r X = \sum_j \big[(r+1)\mathcal{D}[a_j]X + r\mathcal{D}[a_j^*]X\big], \qquad \mathcal{D}[a]X = aXa^* - \tfrac{1}{2}\{a^*a, X\}. \tag{5.5}$$

It fixes τ_r. Set $\delta_s = \mathcal{L}_{r_s}\rho_s$, and let Φ_i be the single-input channel dual to T_i. Then

$$\delta_0 = (1-\theta)\big[\Phi_A(\delta_A) + \Phi_B(\delta_B)\big]. \tag{5.6}$$

Thus relaxing both inputs at their reference occupations gives exactly the output relaxation at r_0.

The coefficient check explains why. In characteristic coordinates the generator has a radial drift and isotropic diffusion proportional to $r + \frac{1}{2}$. The drift is carried to the retained drift by the linear transformation. Each input's diffusion contributes $\lambda_s(r_s + \frac{1}{2})$, including the creation port, whose quadrature reversal preserves Euclidean length. The total diffusion is correct because

$$\sum_{s=A,B,D} \lambda_s(r_s + \tfrac{1}{2}) = r_0 + \tfrac{1}{2}.$$

This identity uses *both* the occupation balance and $\lambda_A - \lambda_B + \lambda_D = 1$. The thermal auxiliary and replacement states are stationary, which leaves (5.6).

The calculation is first made on a number core or on Weyl tests. For $\mathcal{B}_p = \{X : W^p X W^p \text{ is trace class}\}$, ladder estimates give $\mathcal{L}_r : \mathcal{B}_{5/2} \to \mathcal{B}_{3/2}$, and Gaussian channels are continuous at both weights. The fifth moments at the minimum therefore make (5.6) an identity in $\mathcal{B}_{3/2}$, rather than a formal identity of unbounded expressions; see Appendix B.

6 Where the two signs meet

We now have two facts about the same relaxation. Interpolation controls its thermal-balance defect at the output. Covariance lets us insert the relaxation into the stationary equations. To combine them, we need the exact relation between the defect and entropy production.

Normalize entropy production by $\sigma_s = \mathrm{Tr}(\delta_s H_s)/h_s$. Then

$$\sigma_s = \sum_j \|q_{s,j}^{\downarrow}\|^2_{M_s^{\downarrow},*} - (e_s' - n r_s). \tag{6.1}$$

The centered-energy correction is essential: the thermal defect is not entropy production by itself.

Here is the scalar calculation behind (6.1). In an eigenbasis of ρ_s, put

$$v_{lk} = \frac{\log(p_l/p_k)}{h_s}, \qquad K_{lk} = (r_s+1)p_k - r_s p_l.$$

The logarithmic mean gives

$$\frac{K_{lk}^2}{h_s\,\ell(r_s p_l, (r_s+1)p_k)} = (1 - v_{lk})K_{lk}.$$

Consequently the defect norm is $\sum_{j,l,k}(1-v_{lk})K_{lk}|d_{j,lk}|^2$, whereas the dissipator gives $\sigma_s = -\sum_{j,l,k} v_{lk}K_{lk}|d_{j,lk}|^2$. Their difference is

$$\sum_{j,l,k} K_{lk}|d_{j,lk}|^2 = (r_s+1)e_s' - r_s(e_s' + n) = e_s' - n r_s.$$

The canonical extra unit appears once more.

Replacing a_j by d_j changes the generator only by a commutator with ρ_s, whose pairing with H_s vanishes. Restriction to centered tests does not change the dual norm: the diagonal metric weight is p_l, and the defect annihilates I. Finally, $H_s \le CW_s^3$ and the fifth moment make the sums absolutely convergent; the crossed logarithmic terms require at most a fourth moment. Lemma B.3 identifies this matrix calculation with the weighted trace pairing used in stationarity.

Pair (3.9) with δ_i, and use covariance (5.6). The pairing is legitimate because $W_i^{-3/2} H_i W_i^{-3/2}$ is bounded and $W_i^{3/2} \delta_i W_i^{3/2}$ is trace class. Since $\operatorname{Tr} \delta_i = 0$, the result is

$$0 = \sigma_0 - (1+\varepsilon) \sum_i w_i \sigma_i + \varepsilon \sum_i w_i \operatorname{Tr} \delta_i N_i + \zeta P, \qquad P = \sum_i \operatorname{Tr} \delta_i W_i^3. \tag{6.2}$$

The remaining energy terms have simple forms:

$$\operatorname{Tr} \delta_i N_i = -(e_i' - n r_i) - |\mu_i|^2, \tag{6.3}$$

$$e_0' - n r_0 = \sum_i w_i (e_i' - n r_i) + \theta(1-\theta) |\mu_\gamma|^2. \tag{6.4}$$

The second identity uses independence and the signed occupation balance. Substituting (6.1), (6.3) and (6.4) into stationarity cancels every centered-energy term:

$$\begin{aligned} 0 = \| q_0^{\downarrow} \|^2_{M_0^{\downarrow},*} - (1+\varepsilon) A_*^2 - \varepsilon M_*^2 \\ - \theta(1-\theta) |\mu_\gamma|^2 + \zeta P. \end{aligned} \tag{6.5}$$

What disappeared is exactly the occupation correction separating entropy production from the squared defect. Its coefficients cancel because relaxation and the thermal metric use the same ordered pair of rates, reversed at the same creation port. Interpolation now bounds the positive output term by the input defects, while the factor $1+\varepsilon$ makes their negative coefficient strictly larger. The auxiliary thermal slack has become strictness in the final estimate.

The cubic penalty has only a uniformly bounded *upper* drift. Direct calculation gives

$$\mathcal{L}_r^*(W^3) = -3N^3 + 3\big((n+2)r - 1\big)N^2 + \big((9n+6)r - 1\big)N + 7nr.$$

Its leading coefficient is negative, so its supremum over nonnegative integers is finite. Thus $P \le C$, independently of ζ. Combining (6.5) with (5.4) proves

$$A_*^2 + M_*^2 \le \frac{2C\zeta}{\varepsilon} \longrightarrow 0. \tag{6.6}$$

No uniform fifth-moment bound as $\zeta \downarrow 0$ was used.

There is a useful interpretation of the quantity forced to zero. If

$$\mathcal{I}_r(\rho) = -\operatorname{Tr}(\mathcal{L}_r \rho)(\log \rho - \log \tau_r)$$

is thermal relative-entropy production, then (6.1) and the number drift give

$$\frac{\mathcal{I}_r(\rho)}{h(r)} = \sum_j \| q_j^{\downarrow} \|^2_{M^{\downarrow},*} + |\mu|^2.$$

The minimizing inputs are approaching zero total thermal entropy production. To turn that statement into a contradiction, we next turn balance into distance.

7 From thermal balance to trace distance

An exact thermal-balance equation determines the thermal state. A quantitative version should say how far a state can be from thermal equilibrium when that equation is nearly satisfied. The following estimate does so directly in trace norm. It is independent of the amplifier inequality and applies even at the vacuum reference.

Theorem 7.1 (Thermal Stein bound)**.** *Let $n < \infty$, $r \geq 0$, and let ρ be an n-mode finite-energy state. Write τ_r for the product thermal state, with τ_0 the vacuum, and define the trace-class residuals*

$$R_j = (r+1)\rho a_j^* - r a_j^* \rho, \qquad \delta_j = \|R_j\|_1.$$

Then

$$\|\rho - \tau_r\|_1 \leq \pi\kappa_r \sum_{j=1}^{n} \delta_j \leq \pi\sqrt{\frac{n}{r+1}}\left(\sum_{j=1}^{n} \delta_j^2\right)^{1/2}, \tag{7.1}$$

where, for the one-mode thermal state ω_r,

$$\kappa_r = \frac{\mathbb{E}_{\omega_r}\sqrt{N+1}}{r+1} \leq \frac{1}{\sqrt{r+1}}.$$

The norm is the full trace norm. No value of the input energy enters the constant.

The idea is to relax a *test observable*. Let P_t be the Heisenberg thermal attenuator of transmission e^{-t}. Thermal noise smooths its bounded observables:

$$\|[a_j, P_t Z]\|_\infty,\ \|[a_j^*, P_t Z]\|_\infty \leq \kappa_r \sqrt{\frac{e^{-t}}{1-e^{-t}}}\,\|Z\|_\infty.$$

The generator can be written in terms of R_j and R_j^*, so these commutators bound the derivative of $\operatorname{Tr}\rho P_t Z$. Integrating from ρ to equilibrium gives the theorem. The constant π is the integral of the smoothing kernel:

$$\int_0^\infty \sqrt{\frac{e^{-t}}{1-e^{-t}}}\,dt = \pi.$$

The singularity at time zero is integrable.

The smoothing mechanism is inherited from Gao–Rouzé [8, Proposition 6.4, Eqs. (110)–(117) of the preprint]. Appendix D supplies its bounded-observable extension, the normal-slice interpretation, and the finite-energy closure. In particular, the proof does not place an arbitrary finite-energy state in the generator domain.

For use at the variational minimum, we pass from the weighted centered defects to the ordinary residuals. Put $\mu_j = \operatorname{Tr}\rho a_j$ and $\alpha^2 = \sum_j \|q_j^{\downarrow}\|_{M^{\downarrow},*}^2$. If $r > 0$, ρ is faithful and these dual norms are finite, the trace-class operator $Q_{c,j}$ representing $q_j^{\downarrow}$ satisfies

$$\delta_{c,j} := \|Q_{c,j}\|_1 \leq \sqrt{h(r)(r+1)}\,\|q_j^{\downarrow}\|_{M^{\downarrow},*}.$$

Since $R_j = Q_{c,j} + \overline{\mu_j}\rho$, Theorem 7.1 gives

$$\|\rho - \tau_r\|_1 \leq \pi\sqrt{n}\left(\sqrt{h(r)}\,\alpha + \frac{|\mu|}{\sqrt{r+1}}\right). \tag{7.2}$$

Completion of the amplifier proof. Write $\alpha_i^2 = \sum_j \|q_{i,j}^{\downarrow}\|^2_{M_i^{\downarrow},*}$. The weights w_A, w_B are fixed and positive. Hence (6.6) and (7.2) imply, for both minimizing inputs,

$$\|\rho_i - \tau_{r_i}\|_1 = O(\sqrt{\zeta}).$$

This is convergence along the full sequence. The thermal input references produce τ_{r_0}, so continuity of the channel gives the corresponding output convergence.

Trace convergence alone would not suffice for the entropies. The uniform first-energy bounds from Section 3 supply precisely that final step, by Lemma A.1. All three entropies converge, and the entropy portion of (3.7) tends to zero. Its relative-entropy and cubic-moment penalties are nonnegative. Therefore

$$\liminf_{\zeta \downarrow 0} J_\zeta \geq 0,$$

contradicting the fixed negative bound (3.8). This proves Theorem 1.1. □

8 Gaussian equality remembers the shape

Gaussian entropy sees symplectic eigenvalues. An amplifier adds covariance matrices. These are different kinds of information: the first forgets the squeezing frame, while the second is sensitive to how the frames meet. The equality condition records exactly the information the entropy has forgotten.

The Gaussian-input inequality is already recorded in the Gaussian-optimizer literature [10]; a Gaussian proof for the passive counterpart appears in [11, Appendix A.8]. Here we give a spectral refinement and a direct equality classification, independently of Theorem 1.1.

Use interleaved quadratures $R = (Q_1, P_1, \ldots, Q_n, P_n)^T$, with

$$\Omega = \bigoplus_{j=1}^{n} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad T = \bigoplus_{j=1}^{n} \operatorname{diag}(1, -1).$$

The covariance convention is $V_{jk} = \frac{1}{2}\langle\{R_j - d_j, R_k - d_k\}\rangle$, so a physical covariance has symplectic eigenvalues $\nu_j \geq \frac{1}{2}$. Order them increasingly. With V_A, V_B the input covariances, Williamson form writes

$$V = S\left(\bigoplus_{j=1}^{n} \nu_j I_2\right) S^T, \qquad S\Omega S^T = \Omega.$$

The thermal occupations are $\nu_j - \frac{1}{2}$; S records the squeezing and mixing. The amplifier gives

$$W = TV_BT, \qquad V_C = GV_A + KW.$$

The reversal $T\Omega T = -\Omega$ preserves the symplectic eigenvalues of V_B. Displacements are unrestricted and do not affect entropy.

Even one mode shows why shape matters. Take

$$V_A = \tfrac{1}{2} I_2, \qquad W = \tfrac{1}{2} \operatorname{diag}(e^{2t}, e^{-2t}).$$

Both inputs are pure, but for $t \neq 0$,

$$\nu(V_C) = \tfrac{1}{2}\sqrt{(G + Ke^{2t})(G + Ke^{-2t})} > \tfrac{1}{2}(G + K).$$

Thus $S(\rho_C) > g(K)$. The two zero input entropies conceal a mismatch of squeezing. Equality will require the two effective covariance ellipses to have the same normalized shape, although their thermal scales may differ.

For the general statement put

$$\alpha_j = \nu_j(V_A) - \tfrac{1}{2}, \qquad \beta_j = \nu_j(V_B) - \tfrac{1}{2},$$

and define the entropy occupations

$$a = g^{-1}\left(\frac{1}{n}\sum_j g(\alpha_j)\right), \qquad b = g^{-1}\left(\frac{1}{n}\sum_j g(\beta_j)\right).$$

Recall that $S(\rho) = \sum_j g(\nu_j(V) - \frac{1}{2})$.

Theorem 8.1 (Gaussian refinement and equality)**.** *For independent finite-energy n-mode Gaussian inputs,*

$$\frac{1}{n}S(\rho_C) \geq \frac{1}{n}\sum_{j=1}^{n} g\big(G\alpha_j + K(\beta_j + 1)\big) \geq g\big(Ga + K(b+1)\big).$$

Equality between the first and last terms holds if and only if there exist $\nu_A, \nu_B \geq \frac{1}{2}$ and a common positive symmetric symplectic matrix M such that

$$V_A = \nu_A M, \qquad TV_BT = \nu_B M. \tag{8.1}$$

Equivalently, for one real symplectic matrix S,

$$V_A = \nu_A SS^T, \qquad V_B = \nu_B(TST)(TST)^T.$$

Both input displacements are arbitrary. For $n \geq 2$, each input group must have a flat symplectic spectrum, but the two groups may have different occupations. The common S may include squeezing, passive mixing and correlations among modes.

Proof. There are two steps: compare the ordered symplectic spectra, then average their entropy coordinates. Equality in the second step will flatten each input spectrum; equality in the first will then match the covariance shapes.

Spectral comparison. Bhatia–Jain's variational principle [9, Theorem 5(i), Eq. (21)] states that, for a positive definite $2n \times 2n$ matrix A,

$$2\sum_{j=1}^{k} \nu_j(A) = \min_{X^T\Omega X = \Omega_k} \operatorname{tr}(X^T A X), \qquad 1 \leq k \leq n.$$

Apply this to $GV_A + KW$. The minimum of a sum is at least the sum of the minima, so

$$\sum_{j\leq k} \nu_j(V_C) \geq G\sum_{j\leq k} \nu_j(V_A) + K\sum_{j\leq k} \nu_j(V_B).$$

Subtracting $k/2$ gives prefix-sum dominance for the increasing photon vectors. The entropy function transfers this order because

$$g(u) = \int_0^\infty \frac{\min(u,t)}{t(t+1)}\,dt, \qquad \sum_j \min(x_j, t) = \min_{0\leq k\leq n}\left(\sum_{j\leq k} x_j + (n-k)t\right)$$

for increasing $x_1, \ldots, x_n$. Hence

$$S(\rho_C) \geq \sum_j g\big(G\alpha_j + K(\beta_j + 1)\big).$$

Strict averaging in entropy coordinates. The required scalar fact is strict joint convexity, on the closed quadrant, of

$$F(s,t) = g\big(Gg^{-1}(s) + K(g^{-1}(t)+1)\big).$$

A useful way to check it is to expose the small correction to a linear function. For $x > 0$ put $p(x) = x(x+1)g'(x)$. Integrating $g'(x) = \int_0^1 (x+u)^{-1}\,du$ gives the continuous identity

$$p(x) = x + \tfrac{1}{2} - \kappa(x), \qquad \kappa(x) = \int_0^1 \frac{u(1-u)}{x+u}\,du.$$

The function κ is positive and strictly decreasing, with $\kappa(0) = 1/2$. For this calculation write $a = g^{-1}(s)$, $b = g^{-1}(t)$ and $R = Ga + K(b+1)$. The linear parts cancel, leaving

$$\Delta := p(R) - Gp(a) - Kp(b) = G\kappa(a) + K\kappa(b) - \kappa(R) > K\big(\kappa(a) + \kappa(b)\big) > 0,$$

since $R > a$. This remains true when either occupation is zero. In the open quadrant,

$$\begin{aligned} F_{ss} &= \frac{Gg'(R)}{g'(a)^2}\left(\frac{1}{p(a)} - \frac{G}{p(R)}\right), \\ F_{tt} &= \frac{Kg'(R)}{g'(b)^2}\left(\frac{1}{p(b)} - \frac{K}{p(R)}\right), \\ \det \nabla^2 F &= \frac{GKg'(R)^2\Delta}{g'(a)^2 g'(b)^2 p(a)p(b)p(R)} > 0. \end{aligned}$$

Thus the Hessian is positive definite. On $s = 0, t > 0$, the same formula for F_{tt} gives positive edge curvature; on $t = 0, s > 0$, the formula for F_{ss} does so. Continuity at the origin completes strict convexity on the closed quadrant: a nonconstant segment either enters the open quadrant or lies on one of these strictly convex edges.

Jensen now proves the second inequality. Equality forces all pairs (α_j, β_j) to be identical, so each input spectrum is flat: $\nu_j(V_A) = \nu_A$, $\nu_j(V_B) = \nu_B$.

Recovering the shape. Put $q = G\nu_A + K\nu_B$. The $k = 1$ spectral comparison gives $\nu_j(V_C) \geq q$ for every j. Equality in entropy and strict monotonicity of g force all output symplectic eigenvalues to equal q. Therefore

$$(\det V_A)^{1/(2n)} = \nu_A, \qquad (\det W)^{1/(2n)} = \nu_B, \qquad (\det V_C)^{1/(2n)} = q.$$

Minkowski's determinant inequality for $X = GV_A$, $Y = KW$ is an equality. Its equality condition can be seen directly. If λ_j are the eigenvalues of $X^{-1/2}YX^{-1/2}$, Jensen's inequality for the strictly convex function $z \mapsto \log(1 + e^z)$ gives

$$\frac{1}{2n}\sum_{j=1}^{2n} \log(1+\lambda_j) \geq \log\left(1 + \prod_{j=1}^{2n} \lambda_j^{1/(2n)}\right).$$

Equality requires all λ_j to agree. Thus X, Y are proportional and $V_A/\nu_A = W/\nu_B = M$. Williamson form makes M positive symmetric symplectic.

Conversely, (8.1) gives $V_C = qM$, whose symplectic eigenvalues are all q; both inequalities are equalities. This proves both directions, including the pure boundary. □

For several modes, equality requires two agreements: constant occupations within each input group, and matching effective covariance shapes between groups. Different thermal occupations and correlations within a group are both compatible with equality.

9 Where the output entropy goes

The squeezer is unitary on its two output groups C, D. It preserves their joint entropy, even though each marginal entropy may increase. The difference is correlation. Applying the amplifier bound to both ports gives an input-entropy benchmark for how much correlation must be created.

Corollary 9.1. *Let C, D be the two outputs of the gain-G squeezer, and put $a = \mathcal{N}_n(\rho_A)$, $b = \mathcal{N}_n(\rho_B)$. For independent finite-energy inputs,*

$$I(C:D) \geq n\big[g(Ga + K(b+1)) + g(Gb + K(a+1)) - g(a) - g(b)\big].$$

The thermal pair $\rho_A = \tau_a, \rho_B = \tau_b$ attains the bound. In particular, independent pure inputs generate at least $n\,g(K)$ entropy of entanglement.

Proof. Apply Theorem 1.1 to each marginal and use $S(CD) = S(A) + S(B)$, by independence and unitarity. For pure inputs the joint output is pure, so either marginal entropy is its entropy of entanglement. Vacuum inputs attain the stated minimum. □

The thermal calculation already gives the value on the right. The content of the corollary is that arbitrary independent inputs with those entropies cannot reduce it. For pure inputs, the familiar product of two-mode squeezed vacua gives the minimum: additional structure in a pure input cannot lower the entanglement produced at this gain. This is a two-marginal consequence of the entropy inequality, rather than a separate channel-optimization theorem or a claim of priority for the Gaussian case.

Independence is indispensable, even among separable Gaussian states. Let z be a circular complex Gaussian with $\mathbb{E}|z|^2 = v > 0$, and prepare a mixture of coherent pairs with amplitudes

$$\sqrt{K}\,z, \qquad -\sqrt{G}\,\overline{z}.$$

Their retained displacements cancel exactly. Every pair has the same retained covariance, so the output is τ_K, while the input marginals are thermal with means Kv and Gv. Substituting these marginals into the independent-input formula would give $K + 2GKv > K$. Classical correlation between the ports has already defeated that formula; separability cannot replace independence.

The Gaussian equality classification leaves general non-Gaussian equality open. The variational proof excludes strict violations by forcing its own minima toward thermal references; it does not classify arbitrary equality inputs. A quantitative EPnI stability estimate would also need a link from entropy deficit to balance residual, which the thermal Stein estimate alone does not provide.

A Analytic foundations of cubic regularization

The regularized minimum must provide more than a state at which the objective is smallest: it must also justify the unbounded operators that enter the proof. We establish those facts here. The energy-cutoff, Gibbs-majorant and entropy-Hessian framework follows OpenAI's beam-splitter argument [2]; we give the active Gaussian domain estimate and the cubic version used in the main text.

Throughout this appendix, the mode counts and Gaussian parameters are finite and fixed. Constants may depend on them and, at a regularized minimum, on the fixed positive regularization parameter. We assert no uniform high-moment bound as that parameter tends to zero.

Lemma A.1 (Energy compactness and entropy continuity)**.** *On the n-mode Fock space, put $N = \sum_j a_j^* a_j$ and $W = 1 + N$. For every $E < \infty$, the states with $\operatorname{Tr} \rho N \le E$ form a compact set in trace norm, and entropy is continuous on this set. Normalized number cutoffs converge in trace norm and entropy, with bounded energy.*

Proof. Let $P_K = \mathbf{1}_{N \le K}$ and $q = \operatorname{Tr} \rho(I - P_K) \le E/(K+1)$. Hilbert–Schmidt factorization bounds the two off-diagonal blocks together in trace norm by $2\sqrt{q}$. The energy ball is therefore uniformly approximable by finite-dimensional corners. It is closed because number expectation is lower semicontinuous, and hence it is compact.

To control entropy, use the thermal variational bound $S(\omega) \le n g(\operatorname{Tr} \omega N/n)$. Pinching into P_K and its complement increases entropy by a number in $[0, h_2(q)]$, where h_2 is the binary entropy; this follows from the binary flag isometry and the Araki–Lieb inequality. The normalized tail has weighted entropy at most $qng(E/(nq))$, interpreted as zero when $q = 0$. Thus the difference between $S(\rho)$ and the entropy of its low corner, weighted by $1 - q$, is at most

$$h_2(q) + qng(E/(nq)),$$

uniformly tending to zero as $K \to \infty$. The weighted low-corner entropy is a continuous finite-dimensional function, including at zero mass. This proves entropy continuity on the energy ball.

Normalizing the low corner changes its energy by at most the factor $(1-q)^{-1}$. Its entropy has the same limit, since $q \log \operatorname{rank} P_K = O(K^{-1} \log K)$. □

Lemma A.2 (Gaussian graph and weighted-channel bounds)**.** *Let U be a finite composition of passive rotations and finite two-mode squeezers on q modes, with*

$$U^* a_i U = \sum_l (\alpha_{il} a_l + \beta_{il} a_l^*).$$

For every integer $k \ge 0$, both U and U^* are bounded on the graph space $D(W^{k/2})$. If $L = \max_i \sum_l (|\alpha_{il}| + |\beta_{il}|)$, one admissible bound is

$$\|W^{k/2} U \psi\|^2 \le C_{q,k,L}^2 \|W^{k/2} \psi\|^2, \qquad C_{q,k,L}^2 = 1 + \sum_{j=1}^{k} \left\{ {k+1 \atop j+1} \right\} q^j L^{2j} j^j.$$

For $p \in \frac{1}{2}\mathbb{N}_0$, set

$$\mathcal{B}_p = \{X : W^p X W^p \in \mathcal{T}_1\}, \qquad \|X\|_p = \|W^p X W^p\|_1.$$

Then

$$\|\operatorname{Tr}_{\mathrm{env}}(UXU^*)\|_p \le C_p \|X\|_p.$$

Proof. On total occupation M, the sum of squared norms of all annihilator words of length j equals $M(M-1)\cdots(M-j+1)$. The Stirling identity therefore expresses $\|W^{k/2}\psi\|^2$ as a sum of these word norms with coefficients $\left\{ {k+1 \atop j+1} \right\}$. A mixed ladder word of length j has a fixed occupation shift, injective on its nonzero images, and squared coefficient at most $(M+j)^j \le j^j (M+1)^j$. Expanding each transformed word, whose coefficient absolute sum is at most L^j, gives the stated estimate.

The domain argument comes before applying this calculation to a full composition. On finite-particle vectors, an elementary squeezer has images that are finite ladder polynomials applied to its vacuum series

$$\operatorname{sech}(t) \sum_{m \ge 0} (\pm \tanh t)^m |m, m\rangle.$$

These images have every number moment. Graph-norm approximation and closedness of $W^{k/2}$ extend the estimate to the full domain; passive factors preserve that domain. Apply this reasoning to each factor first, then expand the Bogoliubov coefficients of the composition and close the resulting estimate. This proves the asserted graph-space bounds.

For the channel estimate, write $X = W_{\mathrm{tot}}^{-p} T W_{\mathrm{tot}}^{-p}$. The operator $K_p = (W_0^p \otimes I) U W_{\mathrm{tot}}^{-p}$ is bounded, since it factors as

$$K_p = \big[(W_0^p \otimes I) W_{\mathrm{tot}}^{-p}\big]\big[W_{\mathrm{tot}}^{p} U W_{\mathrm{tot}}^{-p}\big].$$

The first factor is a contraction in the number basis; the second is bounded by the graph estimate with $k = 2p$. Partial-trace contractivity applied to $K_p T K_p^*$ proves the weighted bound. In particular, the channel preserves finite fifth moments. For product inputs, $W_{\mathrm{tot}}^s \le \prod_i W_i^s$ in the common number basis, so finite fifth moments of each factor suffice. □

Lemma A.3 (Gibbs regularity and a moment gain)**.** *Let $m \ge 1$ be an integer, $k > 0$, and $B \ge 0$ bounded. The closed form*

$$q_K(\psi) = k\|W^{m/2}\psi\|^2 + \|B^{1/2} W^{1/2}\psi\|^2, \qquad D(q_K) = D(W^{m/2}),$$

defines a positive self-adjoint operator K with compact inverse. Its Gibbs state $\omega = e^{-K}/\operatorname{Tr} e^{-K}$ is faithful, has $\operatorname{Tr} \omega W^{2m-1} < \infty$, and uniquely minimizes $\operatorname{Tr} \sigma K - S(\sigma)$ over states with finite mth number moment.

Proof. Put $C = kI + W^{(1-m)/2} B W^{(1-m)/2}$. Then $kI \le C \le (k + \|B\|)I$ and $q_K(\psi) = \|C^{1/2} W^{m/2}\psi\|^2$. This proves closedness and gives

$$K^{-1} = W^{-m/2} C^{-1} W^{-m/2}.$$

The inverse is compact. By the min–max principle,

$$\#\{j : \nu_j \le R\} \le \operatorname{rank} \mathbf{1}_{W^m \le R/k},$$

and the right-hand side grows polynomially. It follows that $\sum_j e^{-\nu_j}(1+\nu_j)^a < \infty$ for every fixed $a \ge 0$.

Let $K\psi = \nu\psi$ with $\|\psi\| = 1$. Then $\psi \in D(W^{m/2})$. Test the form eigenrelation against each number vector and multiply the coordinate equation by $W^{-1/2}$ to obtain

$$k W^{m-1/2}\psi = \nu W^{-1/2}\psi - B W^{1/2}\psi.$$

The right-hand side is square summable, so this coordinate identity also establishes the required domain membership. Moreover,

$$\|W^{m-1/2}\psi\| \le k^{-1}\big(\nu + \|B\|\sqrt{\nu/k}\big),$$

because $\|W^{m/2}\psi\|^2 \le \nu/k$. Summing the squared norms with Gibbs weights proves the stated moment bound. The argument is in weak form; it does not identify $D(K)$ with the domain of a formal operator sum.

The upper form bound $K \le (k + \|B\|)W^m$ makes all terms in

$$\operatorname{Tr} \sigma K - S(\sigma) = D(\sigma\|\omega) - \log \operatorname{Tr} e^{-K}$$

finite on the stated class. Relative-entropy nonnegativity and its equality case prove uniqueness. The Gibbs state itself belongs to that class because $2m - 1 \ge m$. □

We apply this lemma with $m = 3$. Interpret $\zeta \operatorname{Tr} \rho_i W_i^3$ as $+\infty$ outside its moment domain. For fixed positive parameters, expansion of the thermal relative entropy writes the objective, up to a fixed constant, as

$$\frac{S(\rho_0)}{h_0} - \sum_i \frac{(1+\varepsilon)w_i}{h_i} S(\rho_i) + \varepsilon \sum_i w_i \operatorname{Tr} \rho_i N_i + \zeta \sum_i \operatorname{Tr} \rho_i W_i^3.$$

The bound $S(\rho_i) \le n g(\operatorname{Tr} \rho_i N_i / n) = O(\log(1 + \operatorname{Tr} \rho_i N_i))$ shows that each upper sublevel has bounded input energy independently of ζ, and bounded cubic moment at each fixed $\zeta > 0$. The preceding lemmas, entropy continuity and lower semicontinuity of positive number moments give an attained minimum. A fixed finite-support negative witness gives a negative upper bound independent of sufficiently small ζ.

At a minimum, $\rho_0 \ge \theta \tau_{r_0}$ implies $0 \le H_0 = -\log \rho_0 \le C W_0$ as forms, by operator monotonicity of the logarithm. Increasing bounded spectral truncations define the positive slice $T_i H_0$, and the energy graph bound gives $0 \le T_i H_0 \le C_i W_i$. Replacing output entropy by its supporting cross entropy against H_0 majorizes the objective and agrees with it at the minimum. Thus each input minimizes a Gibbs functional with form Hamiltonian

$$\begin{aligned} K_i &= \frac{h_i}{(1+\varepsilon)w_i} \left[\frac{1-\theta}{h_0} T_i H_0 + \varepsilon w_i N_i + \zeta W_i^3 \right] \\ &= k_i W_i^3 + W_i^{1/2} B_i W_i^{1/2}, \qquad B_i \ge 0 \text{ bounded.} \end{aligned}$$

The lemma identifies $\rho_i = e^{-K_i} / \operatorname{Tr} e^{-K_i}$. Every input is therefore faithful, has finite fifth moment, and satisfies $0 \le H_i \le C_i W_i^3$. Its exact form identity is

$$\frac{(1+\varepsilon)w_i}{h_i} H_i = \frac{1-\theta}{h_0} T_i H_0 + \varepsilon w_i N_i + \zeta W_i^3 + c_i I.$$

The graph bound gives a finite fifth moment for the output as well.

Lemma A.4 (Hessian duality and density)**.** *A centered finite-eigenblock test means a finite matrix block minus its state mean times I; it may have a constant identity tail. At such a minimum, define*

$$\ell(a,b) = \frac{a-b}{\log a - \log b}, \qquad \ell(a,a) = a, \qquad \|Z\|_s^2 = h_s \sum_{l,k} \ell(p_{s,l}, p_{s,k}) |Z_{lk}|^2.$$

For bounded centered tests, with $L_i Z = (T_i Z - \operatorname{Tr}(\gamma Z) I)/\sqrt{\lambda_i}$ and $w_i = (1-\theta)\lambda_i$, one has

$$\|L_i Z\|_i^2 \le \frac{1+\varepsilon}{1-\theta} \|Z\|_0^2.$$

Proof. The exact logarithm identity cancels the linear terms in a one-input variation and gives

$$J_\zeta(\widetilde{\rho}_i) - J_\zeta(\rho_i) = \frac{(1+\varepsilon)w_i}{h_i} D(\widetilde{\rho}_i \| \rho_i) - \frac{1}{h_0} D(\widetilde{\rho}_0 \| \rho_0) \ge 0.$$

A Hermitian traceless finite eigenblock A is an admissible two-sided direction. The state eigenvectors spanning its block belong to $D(W^{5/2})$, so it has finite cubic form moment; faithfulness gives positivity for small real parameters. The input Hessian is

$$Q_i(A) = \sum_{l,k} \frac{|A_{lk}|^2}{\ell(p_{i,l}, p_{i,k})}.$$

For a bounded centered finite-eigenblock test Z, use the Gibbs dual bound

$$D(\omega\|\rho_0) \geq t \operatorname{Tr} \omega Z - \log \operatorname{Tr} e^{\log \rho_0 + tZ}.$$

Only the bounded-test log-partition function is expanded. Its second derivative is

$$\sum_{l,k} \ell(p_{0,l}, p_{0,k}) |Z_{lk}|^2.$$

Optimizing the resulting quadratic bound over A gives

$$(1-\theta)^2 \|T_i Z - \operatorname{Tr}(\gamma Z) I\|_i^2 \leq (1+\varepsilon) w_i \|Z\|_0^2.$$

This step takes no derivative of output entropy along an unbounded direction.

To justify the optimization on the full trace-zero tangent space, observe that trace is continuous in the Q_i norm:

$$|\operatorname{Tr} A|^2 \leq \sum_l \frac{|A_{ll}|^2}{p_{i,l}} \leq Q_i(A).$$

Finite eigenblock truncations, followed by subtraction of their trace in one fixed diagonal entry, are therefore dense in that space. Hermitian and skew-Hermitian decomposition gives the result for complex tests.

Finally, bounded spectral compressions converge strongly with their adjoints, as do their unital positive slices. Dominated convergence in the two arithmetic state norms, together with $\ell(a,b) \leq (a+b)/2$, identifies the completed maps with the actual centered slices on bounded observables. Dividing by λ_i gives the asserted constant. □

B Cubic regularization: generators, weak defects, and entropy production

This appendix establishes the domain statements needed for the cubic penalty $\zeta \operatorname{Tr} \rho W^3$. All mode counts are finite. With $W = 1 + N$, write

$$\mathcal{B}_p = \{X : W^p X W^p \in \mathfrak{T}_1\}, \qquad \|X\|_p = \|W^p X W^p\|_1 \quad (p \geq 0).$$

For a positive state, $\rho \in \mathcal{B}_p$ means $\operatorname{Tr} \rho W^{2p} < \infty$. The preceding Gibbs regularity lemma gives, at each fixed $\zeta > 0$,

$$\operatorname{Tr} \rho_i W_i^5 < \infty, \qquad 0 \leq H_i = -\log \rho_i \leq C_i W_i^3. \tag{B.1}$$

The constants may depend on ζ. The output replacement gives the stronger bound

$$H_0 \leq h_0 N_0 + n \log(r_0 + 1) - \log \theta,$$

because $\rho_0 \geq \theta \tau_{r_0}$ and the logarithm is operator monotone. The finite-mode Bogoliubov graph estimate transfers fifth moments to the joint output and its retained marginal. No fifth-moment bound uniform in $\zeta \downarrow 0$ is required.

B.1 Weighted generator and observable pairings

Lemma B.1. *For every real $p \geq 0$ and fixed $r \geq 0$, the thermal generator*

$$\mathcal{L}_r X = \sum_{j=1}^{n} \big[(r+1)\mathcal{D}[a_j]X + r\mathcal{D}[a_j^*]X\big], \qquad \mathcal{D}[a]X = aXa^* - \tfrac{1}{2}\{a^*a, X\},$$

extends boundedly from $\mathcal{B}_{p+1}$ to $\mathcal{B}_p$. Consequently, under (B.1), *$\delta_i = \mathcal{L}_{r_i}\rho_i \in \mathcal{B}_{3/2}$ and $\operatorname{Tr} \delta_i = 0$.*

Proof. The number-basis ladder relations show that

$$W^p a_j W^{-(p+1)}, \quad W^p a_j^* W^{-(p+1)}, \quad W^p a_j^* a_j W^{-(p+1)}, \quad W^p a_j a_j^* W^{-(p+1)}$$

are bounded. For example, on a number vector of total occupation M, the first two coefficients are bounded by $\sqrt{m_j} M^p/(M+1)^{p+1}$ and $\sqrt{m_j+1}(M+2)^p/(M+1)^{p+1}$, respectively.

Write $X = W^{-(p+1)} T W^{-(p+1)}$. Each weighted jump term then factors as a product of $T \in \mathfrak{T}_1$ with two bounded ladder factors. Each anticommutator term uses a bounded number factor on one side and W^{-1} on the other. This gives the required norm bound on finite number cutoffs. Since those cutoffs commute with W, they are dense in $\mathcal{B}_{p+1}$, and the bound extends by continuity. Their generators have trace zero; convergence in $\mathcal{B}_p$ implies trace-norm convergence, so the trace-zero identity also passes to the limit. □

For $X = X^* \in \mathcal{B}_{3/2}$ and $0 \le H \le CW^3$ as forms, define

$$\operatorname{Tr} XH := \operatorname{Tr}\big[(W^{3/2} X W^{3/2})(W^{-3/2} H W^{-3/2})\big]. \tag{B.2}$$

The second factor is bounded. To justify slicing an unbounded observable, first decompose the weighted operator:

$$Y = W^{3/2} X W^{3/2} = Y_+ - Y_-, \qquad X_\pm = W^{-3/2} Y_\pm W^{-3/2}.$$

Then $X = X_+ - X_-$ and

$$\operatorname{Tr} X_\pm W^3 = \operatorname{Tr} Y_\pm < \infty.$$

Here the positive form trace on the left means $\operatorname{Tr}(W^{3/2} X_\pm W^{3/2})$. The operators $Y_\pm$ are the positive parts of the weighted operator; the resulting $X_\pm$ need not be the positive and negative parts of X itself.

Let Φ_i be the single-input channel with all other ports fixed, and let T_i be its normal positive dual slice. The output energy bound gives $0 \le T_i H_0 \le C_i W_i$. Apply bounded-observable duality to $H_0 \wedge M$ and each positive operator $X_\pm$. Monotone convergence, followed by subtraction, gives

$$\operatorname{Tr} \Phi_i(X) H_0 = \operatorname{Tr} X T_i H_0. \tag{B.3}$$

The energy bound makes all four positive pairings finite, so this subtraction is legitimate.

B.2 Covariance of the thermal generator

For the amplifier followed by the auxiliary attenuator, the retained annihilators are

$$c_j = \sqrt{\lambda_A}\, a_j + \sqrt{\lambda_B}\, b_j^* + \sqrt{\lambda_D}\, d_j, \qquad \lambda_A - \lambda_B + \lambda_D = 1.$$

The auxiliary input D is in τ_{r_D}, and the thermal occupation balance is

$$r_0 = \lambda_A r_A + \lambda_B (r_B + 1) + \lambda_D r_D. \tag{B.4}$$

Write γ for the retained state and $\rho_0 = (1-\theta)\gamma + \theta \tau_{r_0}$.

Lemma B.2. *Under* (B.1)*,*

$$\mathcal{L}_{r_0} \rho_0 = (1-\theta)\big[\Phi_A(\mathcal{L}_{r_A}\rho_A) + \Phi_B(\mathcal{L}_{r_B}\rho_B)\big] \quad \text{in } \mathcal{B}_{3/2}. \tag{B.5}$$

In particular,

$$\operatorname{Tr} \delta_0 H_0 = (1-\theta) \sum_{i=A,B} \operatorname{Tr} \delta_i T_i H_0.$$

Proof. Use real quadratures $R = (Q_1, P_1, \dots, Q_n, P_n)$ with $[Q_j, P_k] = i\delta_{jk}$, and set $W(z) = e^{iz^T R}$. For $\chi_X(z) = \operatorname{Tr} XW(z)$, direct ladder algebra gives

$$\chi_{\mathcal{L}_r X}(z) = -\tfrac{1}{2} z \cdot \nabla \chi_X(z) - \tfrac{1}{2}(r + \tfrac{1}{2})|z|^2 \chi_X(z). \tag{B.6}$$

This is first checked on finite number matrices. Equivalently, the corresponding Gaussian relaxation acts on characteristic functions by

$$\chi(z) \longmapsto \chi(e^{-t/2} z) \exp\bigl[-(r + \tfrac{1}{2})(1 - e^{-t})|z|^2/2\bigr].$$

Finite number matrices are dense in $\mathcal{B}_{5/2}$, and $\mathcal{L}_r : \mathcal{B}_{5/2} \to \mathcal{B}_{3/2}$ is bounded. The derivative term in (B.6) is also continuous under this convergence, because quadratures are bounded relative to $W^{1/2}$. Thus the weak identity holds for every $Y \in \mathcal{B}_{5/2}$. It applies, in particular, to the Gaussian output of a finite number matrix: the graph bound gives that output a fifth moment, although its Fock support is generally infinite.

Put $J = \operatorname{diag}(1, -1, \dots, 1, -1)$. For the product input $\rho_A \otimes \rho_B \otimes \tau_{r_D}$, the output characteristic function is the joint input characteristic function restricted to

$$(z_A, z_B, z_D) = \bigl(\sqrt{\lambda_A} z, \sqrt{\lambda_B} J z, \sqrt{\lambda_D} z\bigr).$$

This restriction formula holds for every joint input operator. It therefore remains applicable to total-number cutoffs, which need not be products. On the restricted characteristic function, the sum of the three drift terms in (B.6) is $-z \cdot \nabla/2$. The diffusion coefficient is

$$\sum_{s=A,B,D} \lambda_s (r_s + \tfrac{1}{2}) = r_0 + \tfrac{1}{2},$$

by (B.4) and the signed canonical relation. This proves covariance on the finite number core when tested against all Weyl operators.

The full product input has finite fifth moment of total number, since its commuting number variables satisfy

$$1 + \sum_s N_s \le \prod_s (1 + N_s).$$

Its finite total-number cutoffs converge in $\mathcal{B}_{5/2}$. The generator bound sends these cutoffs to convergent expressions in $\mathcal{B}_{3/2}$, while the Gaussian graph bound makes the full unitary and partial trace continuous at both weights. Covariance therefore extends in $\mathcal{B}_{3/2}$. Weyl operators separate trace-class operators, so the weak calculation identifies the limits. If the thermal input is also cut off, its cutoff generator must be retained until this limit is taken; its limit is zero.

Finally, $\mathcal{L}_{r_D} \tau_{r_D} = 0$ and $\mathcal{L}_{r_0} \tau_{r_0} = 0$, giving (B.5). Pairing with H_0 is continuous in $\mathcal{B}_{3/2}$ by (B.2); (B.3) then gives the final identity. □

This calculation uses a complete canonical Gaussian transformation. The invariance of the friction part can also be checked directly from $\sum_j (\mathcal{D}[a_j] - \mathcal{D}[a_j^*])$, summed over all joint modes. The diffusion part of a general Bogoliubov channel need not be isotropic. The scalar balance above applies to the stated annihilation and creation ports; it does not provide a substitution rule for arbitrary Gaussian channels.

B.3 Weak convolution and the complex adjoint

For a finite-energy state ρ_s, let $\mu_{s,j} = \operatorname{Tr} \rho_s a_{s,j}$ and $d_{s,j} = a_{s,j} - \mu_{s,j} I$. Define

$$q^{\downarrow}_{s,j}(Z) = (r_s + 1) \operatorname{Tr} \rho_s d^*_{s,j} Z - r_s \operatorname{Tr} \rho_s Z d^*_{s,j},$$
$$q^{\uparrow}_{s,j}(Z) = r_s \operatorname{Tr} \rho_s d_{s,j} Z - (r_s + 1) \operatorname{Tr} \rho_s Z d_{s,j}.$$

Hilbert–Schmidt factorization defines both traces for bounded Z. The functionals vanish on I. Taking adjoints gives the complex-linear identity

$$q^{\uparrow}_{s,j}(Z) = -\overline{q^{\downarrow}_{s,j}(Z^*)}. \tag{B.7}$$

The complex conjugation is essential.

Set

$$L_i Z = \frac{T_i Z - \mathrm{Tr}(\gamma Z) I}{\sqrt{\lambda_i}}, \qquad w_i = (1-\theta)\lambda_i.$$

Pull a retained bounded observable through the Gaussian unitary. The inverse transform gives retained components $\sqrt{\lambda_A} c_j^*$ in a_j^*, $-\sqrt{\lambda_B} c_j^*$ in b_j, and $\sqrt{\lambda_D} c_j^*$ in d_j^*. After the corresponding input defects have been subtracted, the remaining weak commutator has coefficient

$$\begin{aligned}
&\lambda_A(r_0 - r_A) - \lambda_B(r_0 + r_B + 1) + \lambda_D(r_0 - r_D) \\
&\quad = r_0(\lambda_A - \lambda_B + \lambda_D) - [\lambda_A r_A + \lambda_B(r_B + 1) + \lambda_D r_D] = 0.
\end{aligned}$$

The auxiliary thermal defect vanishes. Thus the unsmoothed defect is

$$q^{\downarrow}_{\gamma,j} = \lambda_A q^{\downarrow}_{A,j} \circ L_A - \lambda_B q^{\uparrow}_{B,j} \circ L_B.$$

For the reference r_0, write the uncentered functional as

$$\widetilde{q}_\sigma(Z) = (r_0 + 1)\,\mathrm{Tr}\,\sigma c_j^* Z - r_0\,\mathrm{Tr}\,\sigma Z c_j^*.$$

Then

$$q_\sigma(Z) = \widetilde{q}_\sigma(Z) - \overline{\mu_{\sigma,j}}\,\mathrm{Tr}\,\sigma Z.$$

Since $\widetilde{q}_{\tau_{r_0}} = 0$ and $\mu_0 = (1-\theta)\mu_\gamma$, linearity before centering gives

$$\begin{aligned}
q^{\downarrow}_{0,j}(Z) = {} & w_A q^{\downarrow}_{A,j}(L_A Z) - w_B q^{\uparrow}_{B,j}(L_B Z) \\
& + \theta(1-\theta)\overline{\mu_{\gamma,j}}\,\mathrm{Tr}[(\gamma - \tau_{r_0})Z].
\end{aligned} \tag{B.8}$$

This determines both the sign and the complex conjugation of the mixture term.

To justify the discarded-mode commutators, let

$$\omega = U(\rho_A \otimes \rho_B \otimes \tau_{r_D})U^*, \qquad \gamma = \mathrm{Tr}_{\mathrm{disc}}\,\omega,$$

be the full joint output and its retained marginal. Compress a discarded annihilator e in its own output number basis:

$$P_K = \mathbf{1}_{N_e \le K}, \qquad e_K = P_K e P_K = e P_K.$$

This compression acts only on the discarded tensor factor, and hence $[e_K, Z \otimes I] = 0$ exactly. Its two Hilbert–Schmidt tails satisfy

$$\begin{aligned}
\|(e - e_K)\sqrt{\omega}\|_2^2 &= \mathrm{Tr}\,\omega N_e \mathbf{1}_{N_e > K}, \\
\|\sqrt{\omega}(e - e_K)\|_2^2 &= \mathrm{Tr}\,\omega(N_e + 1)\mathbf{1}_{N_e \ge K}.
\end{aligned}$$

For e^*, the corresponding identities have endpoints shifted by at most one. Finite output energy makes these tails vanish. Hilbert–Schmidt Cauchy–Schwarz therefore gives convergence of both trace orderings, and their weak commutator remains zero in the limit. A joint cutoff in retained and discarded number need not commute with every retained Z, so it cannot be used here.

B.4 Exact entropy production in logarithmic-mean coordinates

Lemma B.3. *Let ρ be faithful, $H = -\log\rho \le CW^3$, and $\operatorname{Tr}\rho W^5 < \infty$. Fix $r > 0$ and $h = \log(1 + 1/r)$. In a spectral basis $\rho = \operatorname{diag}(p_l)$, define*

$$M^{\downarrow}_{lk} = h\ell(rp_l, (r+1)p_k), \qquad M^{\uparrow}_{lk} = h\ell((r+1)p_l, rp_k),$$

where $\ell(x, y) = (x - y)/(\log x - \log y)$ has its continuous value. On centered observables, use

$$\|Z\|^2_{M^{\downarrow}} = \sum_{l,k} M^{\downarrow}_{lk} |Z_{lk}|^2.$$

The weak defects extend continuously to the completion of this space, and

$$\frac{1}{h}\operatorname{Tr}(\mathcal{L}_r\rho)H = \sum_{j=1}^{n} \|q_j^{\downarrow}\|^2_{M^{\downarrow},*} - (e' - nr), \qquad e' = \operatorname{Tr}\rho N - |\mu|^2. \tag{B.9}$$

Moreover, adjunction is an isometry between the two metrics, and

$$\|q_j^{\uparrow}\|_{M^{\uparrow},*} = \|q_j^{\downarrow}\|_{M^{\downarrow},*}.$$

Proof. Put $v_{lk} = h^{-1}\log(p_l/p_k)$ and $K_{lk} = (r+1)p_k - rp_l$. Direct matrix multiplication gives

$$q_j^{\downarrow}(Z) = \sum_{l,k} K_{lk}\overline{(d_j)_{lk}} Z_{lk}, \qquad M^{\downarrow}_{lk} = \frac{K_{lk}}{1 - v_{lk}},$$

with the latter equality interpreted continuously at $v_{lk} = 1$. The squared dual norm is therefore

$$\|q_j^{\downarrow}\|^2_{M^{\downarrow},*} = \sum_{l,k} (1 - v_{lk}) K_{lk} |(d_j)_{lk}|^2. \tag{B.10}$$

On the diagonal, $M^{\downarrow}_{ll} = p_l$, so I has norm one. Since $q_j(I) = 0$, the representing vector is orthogonal to constants; restricting to centered tests does not change its norm. Also, $M^{\uparrow}_{lk} = M^{\downarrow}_{kl}$. Together with (B.7), this proves the adjoint isometry and the equality of dual norms.

Replacing a_j by d_j changes the generator by a commutator with ρ, whose pairing with a bounded function of ρ is zero. Evaluating the dissipator diagonally in the spectral basis gives

$$\frac{1}{h}\operatorname{Tr}(\mathcal{L}_r\rho)H = -\sum_{j,l,k} v_{lk} K_{lk} |(d_j)_{lk}|^2. \tag{B.11}$$

All these sums are absolutely convergent. To check this explicitly, write $E_l = -\log p_l$. The crossed logarithmic sums are

$$\sum_{l,k} p_k E_l |(d_j)_{lk}|^2 = \operatorname{Tr}\rho d_j^* H d_j,$$
$$\sum_{l,k} p_l E_k |(d_j)_{lk}|^2 = \operatorname{Tr}\rho d_j H d_j^*.$$

The form bound $H \le CW^3$ and the ladder shifts bound each by a constant times $\operatorname{Tr}\rho W^4$. For an uncrossed sum, take a normalized spectral eigenvector u, with H-eigenvalue E_u, and use

$$E_u \le C\langle W^3\rangle_u, \qquad \|d_j u\|^2 + \|d_j^* u\|^2 \le C'(1 + |\mu_j|^2)\langle W\rangle_u,$$
$$\langle W^3\rangle_u \langle W\rangle_u \le \langle W^4\rangle_u.$$

The last inequality follows from the nonnegative covariance of two increasing functions of the same scalar number variable. Thus all four sums arising from

$$|v_{lk}K_{lk}| \le h^{-1}(E_l + E_k)\big[(r+1)p_k + rp_l\big]$$

are finite. The terms without logarithms require only finite energy.

For the dissipator evaluation, use the bounded spectral truncations $H_M = H \wedge M$. The bounded positive operators $W^{-3/2}H_M W^{-3/2}$ increase strongly to $W^{-3/2}HW^{-3/2} \le CI$. Their traces against the positive parts of $W^{3/2}\delta W^{3/2}$ converge by monotone convergence, so their difference converges to (B.2). Hence the spectral evaluation agrees with the trace pairing used in stationarity. The same estimates justify the representing vector and (B.10). Their constants may depend on the fixed state through μ; no uniform high-moment estimate is needed.

Subtracting (B.11) from the sum of (B.10) leaves

$$\sum_{j,l,k} K_{lk}|(d_j)_{lk}|^2 = (r+1)e' - r(e'+n) = e' - nr,$$

because $[d_j, d_j^*] = I$. This proves (B.9). □

B.5 The cubic drift

The cubic penalty pairs with $\delta \in \mathcal{B}_{3/2}$ through (B.2). Its exact scalar Heisenberg drift is

$$\begin{aligned}\mathcal{L}_r^*(W^3) &= (r+1)N\big[(W-1)^3 - W^3\big] + r(N+n)\big[(W+1)^3 - W^3\big] \\ &= -3N^3 + 3((n+2)r - 1)N^2 + ((9n+6)r - 1)N + 7nr.\end{aligned}$$

This polynomial has finite supremum on $N \ge 0$. Consequently,

$$\sum_i \operatorname{Tr} \delta_i W_i^3 \le C$$

with C independent of ζ. The stationarity cancellation therefore uses the same entropy-production identity with the cubic penalty.

C Four endpoint comparisons and the twisted logarithmic mean

The four-weight theorem and its analytic construction below are adapted from OpenAI [2, Section 2]. We retain its finite-parameter test space. The logarithmic-mean formulation in the main text is an algebraic rewriting of this result.

The interpolation principle passes from four elementary quadratic-form comparisons to the weight needed in the proof. At a regularized minimum, that weight identifies the squared norm of a centered thermal-balance defect with normalized entropy production plus a centered-energy correction. We give the comparison argument together with the analytic representation on which it depends.

Set

$$f(x) = \frac{x}{\sinh x}, \qquad b(x) = x\coth x, \qquad f(0) = b(0) = 1. \tag{C.1}$$

Both functions are positive and even on $\mathbb{R}$. On Hilbert spaces $\mathcal{H}_j$, $0 \le j \le k$, fix coordinates in which every weight below acts by scalar multiplication. A positive weight X_j defines the possibly unbounded form

$$Q_{X,j}(Z) = \sum_\alpha X_j(\alpha)|Z_\alpha|^2;$$

fixed positive coordinate measures may be absorbed into the coordinates. Let $\mathcal{T} \subset \mathcal{H}_0$ be a linear test space, let $L_i : \mathcal{T} \to \mathcal{H}_i$ be linear, and let $w_i > 0$ for $1 \le i \le k$. We say that the family $X = (X_j)_{j=0}^k$ *satisfies comparison* if

$$Q_{X,0}(Z) \ge \sum_{i=1}^{k} w_i Q_{X,i}(L_i Z) \qquad (Z \in \mathcal{T}). \tag{C.2}$$

The sums on the right are nonnegative coordinate sums.

Theorem C.1 (Four-weight interpolation)**.** *At every coordinate of every space, let $m > 0$ and $x, y \in \mathbb{R}$ be given. Suppose each $Z \in \mathcal{T}$ is supported on finitely many values of the parameter tuple (m, x, y), and $\mathcal{T}$ is invariant under scalar multiplication by functions of this tuple. If the four families*

$$mf(x)e^{x}, \qquad mf(x)e^{-x}, \qquad mf(y)e^{y}, \qquad mf(y)e^{-y} \tag{C.3}$$

satisfy comparison, then so does

$$F = m\frac{f(x)f(y)}{f(x+y)}. \tag{C.4}$$

Finite parameter support does not require finitely many nonzero coordinates. A centered finite matrix, for example, may have an infinite identity tail whose coordinates all have the same parameter value. The theorem permits this situation, which occurs in our application.

C.1 Operations that preserve comparison

The constructions belong to the parallel-sum and operator-mean calculus of Anderson–Duffin and Kubo–Ando [12, 13]. We give direct proofs for forms, including the limits needed for unbounded weights. Throughout this subsection, auxiliary weights are positive functions of (m, x, y).

Positive linear combinations preserve (C.2), as do pointwise limits of such weights. Indeed, each source test has only finitely many parameter values, so its form converges; Fatou's lemma then gives the required inequality on the target spaces.

Comparison also persists under the *parallel sum*

$$X : Y = (X^{-1} + Y^{-1})^{-1}. \tag{C.5}$$

The coordinatewise variational identity is

$$(X : Y)|z|^2 = \min_{u+v=z} \big(X|u|^2 + Y|v|^2\big), \qquad u = \frac{Y}{X+Y}z, \quad v = \frac{X}{X+Y}z.$$

The source minimizers U, V belong to $\mathcal{T}$ by its invariance. Apply the comparisons for X and Y to these tests. Linearity gives $L_iU + L_iV = L_iZ$, and the coordinatewise minimum on each target space gives comparison for $X : Y$. Target minimizers need not belong to a specified test space.

It follows that comparison is preserved by the logarithmic mean $\ell(X, Y) = (X - Y)/(\log X - \log Y)$ and by $XY/\ell(X, Y)$, with $\ell(X, X) = X$. The relevant identities are

$$\frac{1}{\ell(X,Y)} = \int_0^1 \frac{dt}{(1-t)X + tY}, \qquad \frac{XY}{\ell(X,Y)} = \int_0^1 \frac{dt}{t/X + (1-t)/Y}. \tag{C.6}$$

For the first identity, a positive quadrature followed by inversion is a finite parallel sum: if $q_j > 0$ and $V_j > 0$, then

$$\left(\sum_{j=1}^{M} \frac{q_j}{V_j}\right)^{-1} = (V_1/q_1) : \cdots : (V_M/q_M).$$

Here $V_j = (1-t_j)X + t_jY$ already satisfies comparison. The second identity is a positive average of weighted parallel sums. Pointwise limits give the integrals; zero coefficients may be omitted or obtained by a limit. These steps do not use any rule asserting preservation under multiplication of weights.

Apply the two operations to each opposite-exponent pair in (C.3). Their logarithmic mean is m, so we obtain comparison for

$$m, \qquad ms, \qquad mt, \qquad s = f(x)^2, \quad t = f(y)^2. \tag{C.7}$$

We next construct an auxiliary weight from these three.

C.2 A Stieltjes representation

The function $x \mapsto f(x)^2$ decreases from 1 to 0 on $[0,\infty)$. Define B on $(0,1]$ by

$$B\big(f(x)^2\big) = b(x). \tag{C.8}$$

The role of the auxiliary weight is dictated by the final scalar minimization. The quadratic in (C.18) pairs the four endpoint weights on $\xi \mp \phi$ and $\xi \pm \phi$, then adds $E|\phi|^2$. Writing $E = me > 0$ and $a = b(x) + b(y)$, its minimum is

$$\frac{m}{2}\left(a - \frac{(x-y)^2}{a+2e}\right)|\xi|^2.$$

For $x^2 \neq y^2$, the addition formula gives $F/m = (yb(x) + xb(y))/(x+y)$. For this minimum to equal $F|\xi|^2$, the quadratic construction therefore requires

$$(a+2e)(b(x)-b(y)) = x^2 - y^2, \qquad E = \frac{m(t-s)}{2(B(s)-B(t))}, \qquad s = f(x)^2, \quad t = f(y)^2,$$

where the last equality uses $b(z)^2 = f(z)^2 + z^2$. The final minimization will verify the value in every case. The remaining issue is to derive comparison for this E from the three known weights. Initially, for $0 < s, t < 1$, its reciprocal can be written as

$$\frac{1}{E} = \frac{2}{m}\int_0^1 -B'\big(\theta s + (1-\theta)t\big)\,d\theta, \qquad s = f(x)^2, \quad t = f(y)^2. \tag{C.9}$$

The next lemma makes this a positive continuous weight also when $s = 1$ or $t = 1$. More specifically, a Stieltjes representation of $-B'$ will express $1/E$ as a limit of positive combinations of $1/m$ and reciprocals of $\theta(ms) + (1-\theta)(mt) + rm$, with $r \geq 0$. The parallel-sum construction will then give comparison for E.

Lemma C.2. *The function B extends holomorphically to the upper half-plane and across the positive real axis, is real and strictly decreasing there, and satisfies $B(1) = 1$ and $B'(1) = -1$. There are a constant $c \geq 0$ and a positive measure ν on $[0,\infty)$ such that*

$$-B'(s) = c + \int_{[0,\infty)} \frac{d\nu(r)}{s+r} \quad (s > 0), \qquad \int_{[0,\infty)} \frac{d\nu(r)}{1+r} < \infty. \tag{C.10}$$

In particular, $-B'(s) > 0$ for every $s > 0$.

Proof. For $x > 0$, put $s = f(x)^2 \in (0,1)$. Differentiating $s = f(x)^2$ and $B(s) = b(x)$ gives

$$\frac{ds}{dx} = \frac{2s(1-B)}{x}, \qquad \frac{db}{dx} = \frac{B-s}{x}.$$

Since $B(s) = x \coth x > 1$, we may define

$$R(s) = \frac{1-s}{s(B(s)-1)}.$$

The derivative identities give

$$-B'(s) = \frac{B(s)-s}{2s(B(s)-1)} = \frac{1}{2}\left(\frac{1}{s} + R(s)\right). \tag{C.11}$$

The term $1/s$ is a Stieltjes kernel. We shall obtain the corresponding representation for R by first extending B through the parametrization

$$s = \left(\frac{u}{\sin u}\right)^2, \qquad B(s) = u \cot u. \tag{C.12}$$

For $q > 0$, let $p_*(q) \in (\pi/2, \pi)$ be the unique solution of

$$p_*(q) \tan p_*(q) = -q \tanh q,$$

and set

$$D = \{p + iq : q > 0,\ 0 < p < p_*(q)\}, \qquad \mathcal{Q} = \{z : \operatorname{Re} z > 0,\ \operatorname{Im} z < 0\}.$$

Existence, uniqueness and continuity of p_* follow because $p \tan p$ increases strictly from $-\infty$ to 0 on $(\pi/2, \pi)$: its derivative is $(p + \sin p \cos p)/\cos^2 p > 0$. Thus D is connected. The map $a(u) = \sin u/u$ takes D into $\mathcal{Q}$, since

$$\operatorname{Re} a(p+iq) = \frac{p \sin p \cosh q + q \cos p \sinh q}{p^2+q^2} > 0,$$
$$\operatorname{Im} a(p+iq) = \frac{p \cos p \sinh q - q \sin p \cosh q}{p^2+q^2} < 0.$$

For $p > \pi/2$, the first sign is exactly the boundary equation; for $p \le \pi/2$, it is immediate. For the second sign, use $\tan p/p > 1 > \tanh q/q$ when $p < \pi/2$; the case $p \ge \pi/2$ is immediate.

The map is proper into $\mathcal{Q}$. Its finite boundary pieces map to the coordinate axes: $p = 0$ and $q = 0$ map to the positive real axis, while $p = p_*(q)$ maps to the negative imaginary axis. The limiting corners $u = 0, \pi$ map to $1, 0$, respectively. Moreover,

$$|a(p+iq)| \ge \frac{\sinh q}{\sqrt{\pi^2+q^2}} \longrightarrow \infty \qquad (q \longrightarrow \infty)$$

uniformly for $0 \le p \le \pi$. The inverse image of a compact subset of $\mathcal{Q}$ therefore cannot approach the boundary or infinity. There are no critical points in D, because

$$\operatorname{Im}(u \cot u) = \frac{q \sin(2p) - p \sinh(2q)}{\cosh(2q) - \cos(2p)} < 0, \tag{C.13}$$

whereas $a'(u) = 0$ would imply $u \cot u = 1$. The strict inequality follows from $\sin(2p) \le 2p$ and $\sinh(2q) > 2q$.

Properness and the inverse function theorem show that $a(D)$ is both open and relatively closed in $\mathcal{Q}$, hence equals $\mathcal{Q}$. Each fiber is finite. Local inverse neighborhoods of its points give an evenly covered neighborhood: otherwise, additional inverse points arbitrarily close to the chosen image point would, by properness, accumulate at that fiber. Thus a is a covering map. A connected covering of the simply connected quadrant has one sheet. Indeed,

inverse branches continue along paths through successive evenly covered neighborhoods, and homotopies make their continuation independent of the path. It follows that $a : D \to Q$ has a holomorphic inverse.

For $\operatorname{Im} s > 0$, take the branch of $s^{-1/2}$ in Q and define $u = a^{-1}(s^{-1/2})$. Equations (C.12) and (C.13) define a holomorphic B with $\operatorname{Im} B < 0$. To continue it across a point $s_0 > 0$, the preceding bound on $|a|$ first bounds q as $s \to s_0$ from above. Every cluster point of u lies on $p = 0$ or $q = 0$, since the curved boundary has purely imaginary image. On these boundary pieces, $\sinh q/q$ increases strictly from 1 to ∞, and $\sin p/p$ decreases strictly from 1 to 0 for $0 < p < \pi$. The boundary limit is therefore unique: $u = iq$ if $0 < s_0 < 1$, and $u = p \in (0, \pi)$ if $s_0 > 1$. The derivatives there are nonzero, so local analytic inversion continues B across s_0. At $s_0 = 1$, use u^2 as the parameter:

$$\left(\frac{u}{\sin u}\right)^2 = 1 + \frac{u^2}{3} + O(u^4), \qquad u \cot u = 1 - \frac{u^2}{3} + O(u^4).$$

This gives $B(1) = 1$ and $B'(1) = -1$. The two real parametrizations also show that B is strictly decreasing on $(0, \infty)$ and agrees with (C.8).

We now use the classical Herglotz representation [14]. In the required form, if P is holomorphic with $\operatorname{Im} P \geq 0$ in the upper half-plane and extends holomorphically with real values across $(0, \infty)$, then

$$P(s) = a_0 + \beta s + \int_{(-\infty,0]} \left(\frac{1}{r - s} - \frac{r}{1 + r^2}\right) d\mu(r), \qquad \beta \geq 0, \tag{C.14}$$

where $a_0 \in \mathbb{R}$, $\mu \geq 0$ and $\int (1 + r^2)^{-1} d\mu(r) < \infty$. Analytic continuation and reality on $(0, \infty)$ localize the standard representation to the negative real axis. We use this classical result without repeating its Poisson-kernel proof.

Apply (C.14) to $P = -B$ and subtract the value at 1. The function

$$G(s) = \frac{B(s) - 1}{1 - s} = \beta + \int_{(-\infty,0]} \frac{d\mu(r)}{(1 - r)(s - r)} \tag{C.15}$$

has the removable value $G(1) = 1$. It is positive on the positive real axis. Also, $H(s) = sG(s)$ has nonnegative imaginary part in the upper half-plane, term by term: $\beta \geq 0$ and $\operatorname{Im}(s/(s - r)) \geq 0$ for $r \leq 0$. The function H has no zero there. Otherwise, the nonnegative harmonic function $\operatorname{Im} H$ would attain zero at an interior point; the minimum principle would force H to be a real constant, and its positive real-axis values exclude the constant zero. Hence

$$R(s) = \frac{1}{sG(s)}$$

continues the real function R holomorphically to the upper half-plane and across $(0, \infty)$. It is positive on that interval and satisfies $\operatorname{Im}(-R) \geq 0$.

Apply (C.14) again, this time to $-R$, and denote its coefficients by $\widetilde{\beta}, \widetilde{\mu}$. For real $s > 1$,

$$\begin{aligned} 0 \leq R(1) - R(s) &= \widetilde{\beta}(s - 1) + \int_{(-\infty,0]} \left(\frac{1}{1 - r} - \frac{1}{s - r}\right) d\widetilde{\mu}(r) \\ &\leq R(1). \end{aligned}$$

Thus $\widetilde{\beta} = 0$, and monotone convergence gives $\int (1 - r)^{-1} d\widetilde{\mu}(r) \leq R(1)$. We may therefore write

$$R(s) = c_0 + \int_{(-\infty,0]} \frac{d\widetilde{\mu}(r)}{s - r}, \qquad c_0 = R(1) - \int_{(-\infty,0]} \frac{d\widetilde{\mu}(r)}{1 - r} \geq 0. \tag{C.16}$$

This identity first follows on the positive real axis. Local uniform convergence and analytic continuation extend it to the upper half-plane.

By (C.11), the holomorphic functions $-B'$ and $\frac{1}{2}(1/s + R(s))$ agree on $0 < s < 1$. The identity theorem extends the equality to the upper half-plane and the positive real axis. Replacing r by $-r$ in (C.16) now yields (C.10). The term $1/(2s)$ contributes an atom of mass $1/2$ at zero and gives strict positivity. □

C.3 The additional weight and the final minimization

The lemma extends (C.9) continuously and positively to all $s, t > 0$, including the values in (C.7). The fundamental theorem of calculus gives

$$E = \begin{cases} \dfrac{m(t-s)}{2(B(s)-B(t))}, & s \neq t, \\ -\dfrac{m}{2B'(s)}, & s = t. \end{cases} \tag{C.17}$$

Lemma C.2 places E within the comparison-preserving calculus. After substituting (C.10), each reciprocal term has the form

$$\frac{1}{m(\theta s + (1-\theta)t + r)} = \frac{1}{\theta(ms) + (1-\theta)(mt) + rm},$$

and the constant term is a multiple of $1/m$. Positive finite quadratures followed by inversion are parallel sums of positive multiples of these positive linear combinations of m, ms, mt. Truncating the r-integral and refining the partitions gives pointwise convergence for every $s, t > 0$: the integrands are continuous on each bounded interval, and the tail is controlled by $\int (1+r)^{-1} d\nu(r) < \infty$. The finite-parameter hypothesis on the source and Fatou's lemma on the targets therefore give comparison for E.

We finish the proof of Theorem C.1 by evaluating the scalar minimum. For $\xi, \phi \in \mathbb{C}$, define

$$Q(\xi, \phi) = \frac{1}{4} \sum_{\pm} m f(x) e^{\pm x} |\xi \mp \phi|^2 + \frac{1}{4} \sum_{\pm} m f(y) e^{\pm y} |\xi \pm \phi|^2 + E|\phi|^2. \tag{C.18}$$

Write $a = b(x) + b(y)$ and $e = E/m$. The identity $f(x)e^{\pm x} = b(x) \pm x$ gives

$$Q(\xi, \phi) = \frac{ma}{2}|\xi|^2 + m\left(\frac{a}{2} + e\right)|\phi|^2 - m(x-y)\,\mathrm{Re}(\overline{\xi}\phi).$$

The unique minimizer and its value are

$$\phi = \frac{x-y}{a+2e}\xi, \qquad \min_{\phi} Q(\xi, \phi) = \frac{m}{2}\left(a - \frac{(x-y)^2}{a+2e}\right)|\xi|^2. \tag{C.19}$$

If $x^2 \neq y^2$, the identity $b(z)^2 = f(z)^2 + z^2$ and (C.17) imply

$$(a+2e)(b(x) - b(y)) = x^2 - y^2.$$

The minimum in (C.19) is therefore

$$m\frac{yb(x) + xb(y)}{x+y}|\xi|^2 = m\frac{f(x)f(y)}{f(x+y)}|\xi|^2, \tag{C.20}$$

where the second equality is the addition formula for sinh, with continuous values when x or y is zero.

The cases $x^2 = y^2$ can be evaluated directly. If $y = x$, the minimizer is $\phi = 0$ and the value is $mb(x)|\xi|^2 = mf(x)^2|\xi|^2/f(2x)$. If $y = -x \neq 0$, write $s = f(x)^2$ and $b = b(x)$. Equations (C.17) and (C.11) give

$$e = \frac{s(b-1)}{b-s}, \qquad b + e = \frac{x^2}{b-s}.$$

Then (C.19) gives $m(b - x^2/(b+e))|\xi|^2 = ms|\xi|^2$, as required. At $x = y = 0$, we have $E = m/2$, $\phi = 0$, and the value is $m|\xi|^2$. Thus (C.20) holds in every case.

For a source test Z, let Φ be its coordinatewise minimizer in (C.19). The invariance assumption gives $\Phi \in \mathcal{T}$. Apply the four endpoint comparisons to $Z \mp \Phi$ and $Z \pm \Phi$, with the signs in (C.18), and apply the comparison for E to Φ. By (C.20), their sum on the source is $Q_{F,0}(Z)$. The corresponding target sum satisfies

$$\sum_i w_i \sum_\alpha \mathcal{Q}_i\big((L_i Z)_\alpha, (L_i \Phi)_\alpha\big) \geq \sum_i w_i Q_{F,i}(L_i Z).$$

This proves Theorem C.1.

An exact remainder. No commutation of L_i with parameter multipliers is used in the last step. Completing the square makes this explicit. Label the endpoint weights $X_\pm = mf(x)e^{\pm x}$ and $Y_\pm = mf(y)e^{\pm y}$, and put

$$C = m\left(\frac{b(x) + b(y)}{2} + \frac{E}{m}\right), \qquad \kappa = \frac{x - y}{b(x) + b(y) + 2E/m}.$$

The scalar quadratic satisfies

$$\mathcal{Q}(\xi, \phi) = F|\xi|^2 + C|\phi - \kappa\xi|^2.$$

Writing $\Delta_W(Z) = Q_{W,0}(Z) - \sum_i w_i Q_{W,i}(L_i Z)$ and $\Phi = \kappa_0 Z$, we obtain

$$\begin{aligned}\Delta_F(Z) = {} & \tfrac{1}{4}\big[\Delta_{X_+}(Z - \Phi) + \Delta_{X_-}(Z + \Phi) + \Delta_{Y_+}(Z + \Phi) + \Delta_{Y_-}(Z - \Phi)\big] + \Delta_E(\Phi) \\ & + \sum_i w_i \sum_\alpha C_i(\alpha)\big|(L_i\Phi)_\alpha - \kappa_i(\alpha)(L_i Z)_\alpha\big|^2.\end{aligned}$$

This identity holds whenever the displayed forms are finite, in particular on the finite-parameter tests in the theorem. Every term on the right is nonnegative. Failure of L_i to intertwine the minimizing scalar multiplier therefore contributes a nonnegative remainder.

D Proof of the thermal Stein bound

We prove Theorem 7.1 first on finite Fock matrices and then pass to finite-energy states. The limiting argument uses trace-class convergence of the ladder residuals, so it imposes no generator-domain assumption on the limiting state.

Finite energy makes every required ladder product trace class. For example,

$$a_j\rho = (a_j\rho^{1/2})\rho^{1/2}, \qquad \|a_j\rho^{1/2}\|_2^2 = \operatorname{Tr}\rho N_j < \infty.$$

Hilbert–Schmidt factorization therefore applies; the other products use $a_j a_j^* = N_j + 1$.

The attenuator semigroup and normal slices. Let P_t be the Heisenberg thermal attenuator with transmission $\eta = e^{-t}$ and environment $\tau_r = \omega_r^{\otimes n}$. Its predual P_t^* is a strongly continuous semigroup with generator

$$\mathcal{L}_r\sigma = \sum_j \big((r+1)\mathcal{D}[a_j]\sigma + r\mathcal{D}[a_j^*]\sigma\big), \qquad \mathcal{D}[a]\sigma = a\sigma a^* - \tfrac{1}{2}\{a^*a, \sigma\}.$$

For a finite Fock matrix σ, its characteristic function obeys

$$\chi_{P_t^*\sigma}(\xi) = \chi_\sigma(e^{-t/2}\xi)\exp\big[-\tfrac{1}{2}(r+\tfrac{1}{2})(1-e^{-t})|\xi|^2\big].$$

Direct ladder algebra gives $\partial_t \chi_{P_t^*\sigma} = \chi_{P_t^*\mathcal{L}_r\sigma}$. The operator $\mathcal{L}_r\sigma$ has finite rank, and trace-class strong continuity makes $\int_0^t P_s^*\mathcal{L}_r\sigma\, ds$ a Bochner integral. Integrating the characteristic-function identity and using uniqueness of Weyl characteristic functions yields

$$P_t^*\sigma - \sigma = \int_0^t P_s^*\mathcal{L}_r\sigma\, ds.$$

Strong continuity now gives the trace-norm derivative at zero. Thus finite Fock matrices belong to the generator domain.

The beam splitter is strongly continuous at transmission one and zero. At zero transmission the channel replaces every state by τ_r, and hence, for every state σ,

$$P_t^*\sigma \longrightarrow \sigma \quad (t \downarrow 0), \qquad P_t^*\sigma \longrightarrow \tau_r \quad (t \to \infty) \tag{D.1}$$

in trace norm.

For a bounded joint operator X and a trace-class environment operator A, the notation

$$\mathrm{Tr}_B[(I\otimes A)X] = (\mathrm{id}\otimes\varphi_A)(X), \qquad \varphi_A(Y) = \mathrm{Tr}\, AY,$$

denotes a normal slice. Its norm is at most $\|A\|_1\|X\|_\infty$; the joint product need not be trace class.

Smoothing of bounded observables. Fix $t > 0$ and a bounded observable Z, and set $X = U_\eta^*(Z\otimes I)U_\eta$. In one system–environment pair the retained and discarded annihilators are

$$c = \sqrt{\eta}\, a + \sqrt{1-\eta}\, b, \qquad d = -\sqrt{1-\eta}\, a + \sqrt{\eta}\, b.$$

Write $D_A(z) = e^{za^* - \overline{z}a}$ for a system displacement, and similarly for the environment. Displacing the system input by z and the environment input by $w = \sqrt{\eta/(1-\eta)}\, z$ produces the same retained displacement. Their difference acts only on the discarded mode, so

$$D_A(z)^* X D_A(z) = D_B(w)^* X D_B(w).$$

Taking the thermal slice and moving the environment displacement onto the density gives the exact identity

$$D_A(z)^*(P_tZ)D_A(z) = \mathrm{Tr}_B[(I\otimes D_B(w)\tau_r D_B(w)^*)X].$$

The displacement orbit of the thermal density is trace-norm differentiable, as follows also from its geometrically decaying Fock coefficients. Differentiate with respect to $\overline{z}$ at zero, as a form on finite Fock vectors. The resulting weak commutator has the bounded extension

$$[a_j, P_tZ] = \sqrt{\frac{\eta}{1-\eta}}\ \mathrm{Tr}_B\big[(I\otimes[\tau_r, b_j])X\big].$$

Differentiation with respect to z gives the corresponding statement for a_j^*.

The one-mode commutator $[\omega_r, b]$ is a weighted shift. For $p_m = r^m/(r+1)^{m+1}$, its singular values are $\sqrt{m+1}(p_m - p_{m+1})$. Therefore

$$\|[\omega_r, b]\|_1 = \frac{1}{r+1}\sum_{m\geq 0} p_m\sqrt{m+1} = \kappa_r \leq \frac{1}{\sqrt{r+1}}.$$

At $r = 0$, the thermal state is the vacuum and the only nonzero singular value is one, so $\kappa_0 = 1$. The other environment modes have trace one. Thus both ladder commutators satisfy

$$\|[a_j, P_t Z]\|_\infty,\ \|[a_j^*, P_t Z]\|_\infty \leq \kappa_r\sqrt{\frac{e^{-t}}{1-e^{-t}}}\,\|Z\|_\infty. \tag{D.2}$$

This is the thermal specialization of the smoothing calculation of Gao–Rouzé [8, Proposition 6.4 of the preprint]. The displacement identity above supplies the extension to every bounded Z.

The generator pairing and its integral. Suppose first that ρ has finite Fock support. For the residuals $R_j = (r+1)\rho a_j^* - r a_j^*\rho$, direct expansion gives

$$\mathcal{L}_r\rho = \tfrac{1}{2}\sum_j \big([a_j, R_j] + [R_j^*, a_j^*]\big).$$

Each residual has finite support up to one ladder step, so it can be paired with the bounded weak commutators in (D.2). Writing $\delta_j = \|R_j\|_1$, we obtain, for $\|Z\|_\infty \leq 1$,

$$\begin{aligned}\left|\frac{d}{dt}\operatorname{Tr}\rho P_t Z\right| &= \big|\operatorname{Tr}(\mathcal{L}_r\rho)P_t Z\big| \\ &\leq \kappa_r\sum_j \delta_j\sqrt{\frac{e^{-t}}{1-e^{-t}}}.\end{aligned}$$

The two commutator estimates combine with the prefactor $1/2$. Since

$$\int_0^\infty \sqrt{\frac{e^{-t}}{1-e^{-t}}}\,dt = \int_0^1 \frac{du}{\sqrt{u(1-u)}} = \pi,$$

integration and (D.1) give $|\operatorname{Tr}(\rho - \tau_r)Z| \leq \pi\kappa_r\sum_j \delta_j$. Trace duality, followed by Cauchy–Schwarz, proves (7.1) for finite-support states. The norm convention is the full trace norm, so no additional factor of two appears.

Finite-energy closure and centered norms. Let $P_K = \mathbf{1}_{N\leq K}$ and $\rho_K = P_K\rho P_K/\operatorname{Tr}(P_K\rho)$, choosing K with positive denominator. Put $W = 1 + N$. Finite energy means that $T = W^{1/2}\rho W^{1/2}$ is trace class. Since P_K commutes with W,

$$W^{1/2}\rho_K W^{1/2} = \frac{P_K T P_K}{\operatorname{Tr}(P_K\rho)} \longrightarrow T \quad \text{in trace norm.}$$

Every operator $a_j W^{-1/2}$, $a_j^* W^{-1/2}$ and its adjoint is bounded. Factoring each ladder product around $W^{1/2}(\rho_K - \rho)W^{1/2}$ therefore proves $R_j(\rho_K) \to R_j(\rho)$ in trace norm. Passing to the limit in the finite-support bound proves the theorem for every finite-energy state.

Finally, assume $r > 0$ and faithfulness, and let $q_j^{\downarrow}$ be the centered functional. For any contraction U, put $Z = U - \operatorname{Tr}(\rho U) I$. The metric bound (2.5) and the arithmetic upper bound for the logarithmic mean give

$$\|Z\|_{M^{\downarrow}}^2 \le \frac{h(r)(r+1)}{2}\left(\operatorname{Tr}\rho ZZ^* + \operatorname{Tr}\rho Z^*Z\right) \le h(r)(r+1).$$

Bounded spectral compressions justify using Z in the completed weighted space. Since $q_j^{\downarrow}(I) = 0$, taking the supremum over U yields

$$\delta_{c,j} \le \sqrt{h(r)(r+1)}\,\|q_j^{\downarrow}\|_{M^{\downarrow},*}.$$

If $Q_{c,j}$ represents the centered functional and $\mu_j = \operatorname{Tr}\rho a_j$, then $R_j = Q_{c,j} + \overline{\mu_j}\rho$. This identity and Cauchy–Schwarz give (7.2).

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